

One Step, One Lead: Mitigating Higher-Order Interference in Multi-Domain Reinforcement Learning via Cross-Step Control

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Abstract

Reinforcement learning (RL) across multiple domains can broaden the reasoning capabilities of large language models (LLMs), yet joint training often degrades individual-domain performance and can destabilize optimization. Existing work typically diagnoses such interference from a single-step view using first-order gradient alignment or curvature-based proxies. We show that this view can miss a critical form of sequential interference: same-point domain gradients may remain nearly orthogonal even when consecutive realized updates partially reverse one another in output space. We further show that consecutive token log-probability footprints recover this interaction directly from adjacent checkpoints as a local second-order interaction in output space, without explicitly reconstructing same-step curvature. Building on this insight, we propose OSOL, which designates a focus domain at each iteration, uses the preceding checkpoint footprint to rank token-level rebound risk, and applies a drift-ranked, adaptively scaled correction within the standard GRPO update. Our analysis shows that this correction suppresses the targeted cross-step output backtracking component. Controlled studies further show that cross-step backtracking is more strongly associated with subsequent task damage than same-point gradient diagnostics, while the preceding footprint ranks future rebound risk more accurately than Hessian-based proxies. On Qwen3-30B-A3B, OSOL reaches a domain-macro average of 0.4822, improving by 5.7% over the strongest compared baseline, without explicit higher-order differentiation.

1 Introduction

Reinforcement learning (RL) has been widely used to improve the reasoning capabilities of large language models (LLMs) in individual domains Shao et al. [2024], Lin et al. [2026a], Yu et al. [2026]. However, joint training across heterogeneous domains often yields uneven domain-wise gains and can even destabilize training Cai et al. [2026], Yang et al. [2026b], Li et al. [2025]. The resulting degradation can be strongly domain dependent, as illustrated by the gap between mixed-domain and single-domain GRPO in Figure 1. Such interference limits the ability of RL to improve a broad set of capabilities within a single foundation model. We investigate how mixed-domain updates interact along the training trajectory and how this interference can be controlled without introducing auxiliary expert models or reward signals beyond those already used by the underlying RL training Lin et al. [2026b], Liu et al. [2026].

Explaining this domain-dependent degradation requires identifying which interactions among heterogeneous updates are associated with subsequent capability degradation. Existing approaches mainly analyze and control such interference within a single training step, either by comparing first-order gradients or by estimating higher-order curvature Yu et al. [2020], Liu et al. [2021], Yang et al. [2026a]. However, our controlled study shows that same-point first-order gradient alignment does not reliably reflect the task damage caused by the subsequent update, as shown in Figure 2(a). Nine of 16 replicates have positive gradient cosine (i.e., are aligned), yet aligned cases can still incur substantial task damage. This result indicates that same-point alignment alone is insufficient for diagnosing sequential

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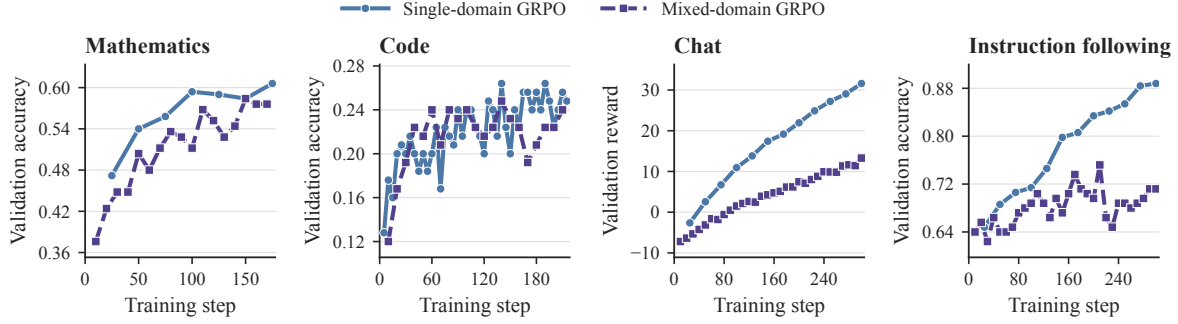


Figure 1: **Mixed-domain GRPO leaves substantial domain-specific headroom relative to isolated training.** Validation trajectories of Qwen3-4B trained with GRPO either separately on each domain or jointly on the four-domain mixture. Clear gaps emerge in Mathematics, Chat, and Instruction Following, with a smaller separation in Code, highlighting the domain dependence of joint-training degradation.

interference. This limitation has motivated higher-order approaches, which use curvature to estimate how one domain update changes the effect of another Liang et al. [2026], Yang et al. [2026a]. However, expanding the parameter coverage of a Hessian proxy yields only modest gains in token-level rebound identification while substantially increasing peak GPU memory, revealing an unfavorable trade-off between estimation quality and computational cost, as shown in Figure 2(b,c). The limited improvement under broader coverage suggests that increasing parameter coverage within a local curvature approximation recovers only limited additional information relative to the realized checkpoint trajectory. Together, these findings motivate us to move beyond gradient and curvature estimates constructed within a single step and instead examine the actual token log-probability footprints left by consecutive updates along the training trajectory.

Building on this cross-step view, we show that adjacent token footprints reveal how consecutive updates interact in task-conditioned output space and provide a principled signal for reducing the resulting cross-step output backtracking. This motivates OSOL, which designates a focus domain at each iteration and converts the preceding token footprint into fine-grained history corrections within the standard GRPO update. These corrections are derived directly from the token-level policy shift left by the preceding update, requiring neither auxiliary supervision nor explicit higher-order derivatives. On Qwen3-30B-A3B, OSOL reaches a domain-macro average of 0.4822, improving by 5.7% over MGS, the strongest baseline, while also outperforming CGPO. The main contributions of this paper are summarized as follows:

- 1) We introduce a cross-step view of higher-order interference in multi-domain policy optimization, shifting the analysis from same-point gradient relations and the curvature sensitivity of isolated parameter displacements to the mixed interaction between consecutive updates in task-conditioned output space. We prove that the inner product of the token log-probability footprints induced by the preceding update and the current base update locally recovers this interaction under the empirical output Fisher, making it observable from adjacent checkpoints without explicitly constructing the Fisher matrix. We further show that opposing token movements give this interaction a direct trajectory interpretation as cross-step output backtracking.

- 2) Building on this analysis, we propose OSOL an online cross-step controller integrated into the standard policy gradient update. At each iteration, it designates a focus domain, forms a normalized task-weighted base coefficient, and computes token-level drift from the log-probability difference of the response tokens across adjacent checkpoints; negative drift defines the intervention set, its rank determines relative penalty strength, and adaptive scaling keeps the residual at a controlled fraction of the current base-coefficient dispersion. The resulting residual is added directly to the objective, and we prove that it contracts the targeted negative-to-positive backtracking component under a local response condition.

- 3) Empirical evaluations validate both the diagnostic and optimization benefits of the cross-step view. In controlled studies, cross-step backtracking mass correlates positively with subsequent task damage ($\rho_s = 0.40$, 95% bootstrap CI [0.05, 0.69]), whereas same-point global and module diagnostics yield correlations of -0.19 and -0.22 , respectively. Checkpoint-footprint ranking also improves rebound ROC-AUC from 0.498 for the Hessian proxy to 0.567. In full multi-domain training on Qwen3-30B-A3B, OSOL improves the domain-macro average by 5.7% over MGS, the strongest compared baseline, while also outperforming curvature-guided CGPO. Ablations further show

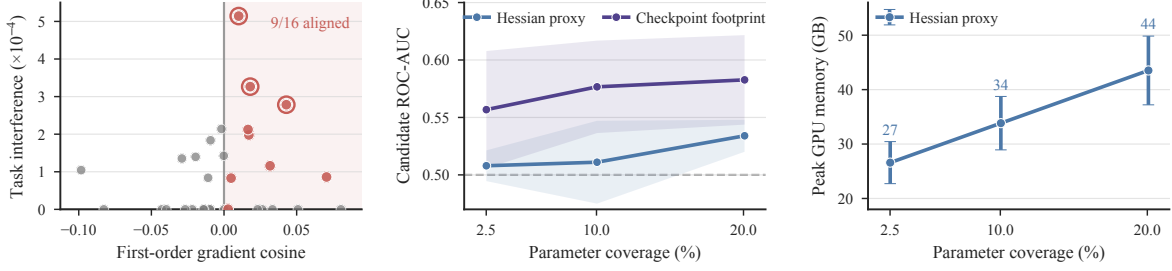


Figure 2: **Same-point alignment misses sequential damage, while broader curvature estimation remains costly.**

(a) Positive same-point gradient cosine does not preclude substantial task damage after the subsequent domain update; circles highlight high-damage aligned cases. (b, c) Expanding Hessian-proxy parameter coverage modestly improves token-level rebound identification but increases peak GPU memory.

that equalizing the focus and non-focus weights or replacing drift-ranked token weights with random weights reduces the domain-macro average by 51.4% and 50.8%, respectively.

2 Related Work

Recent studies of RL post-training across heterogeneous reasoning domains report asymmetric transfer, domain degradation, task-dependent gradient imbalance, and sensitivity to data composition and training order [Cheng et al., 2026, Li et al., 2025, Wu et al., 2026, Yang et al., 2026b]. Existing optimization methods mainly balance task contributions or coordinate gradients evaluated at the same checkpoint. Adaptive weighting methods balance gradient magnitudes or task convergence rates [Chen et al., 2018, Gong et al., 2024], while multi-objective methods seek Pareto-efficient, bargaining-based, or balanced-progress solutions [Sener and Koltun, 2018, Navon et al., 2022, Liu et al., 2023]. PC-Grad and CAGrad instead modify conflicting task gradients before aggregation [Yu et al., 2020, Liu et al., 2021]. Recent LLM-specific RL methods instantiate these ideas through module-level gradient surgery, adaptive task weighting and sampling, and transfer-aware curricula [Cai et al., 2026, Ramesh et al., 2026, Yang et al., 2026c]. These approaches primarily regulate task contributions or same-step gradient interactions, but do not directly model interference between consecutive mixed-domain updates.

Characterizing such cross-step interaction connects three relevant perspectives: second-order sensitivity, output-space policy geometry, and checkpoint-based observability. Influence functions estimate sensitivity to infinitesimal training perturbations through inverse-Hessian information [Koh and Liang, 2017]. Recent local-perturbation theory models cross-domain damage through the curvature of an earlier-domain objective under a later displacement [Yang et al., 2026a], while CGPO uses cross-domain curvature information to shape multi-domain policy updates [Liang et al., 2026]. Natural-gradient and trust-region methods provide a complementary policy-geometry view, where the Fisher metric and policy KL quantify local changes in the action distribution [Amari, 1998, Kakade, 2001, Schulman et al., 2015, Martens, 2020]. Work on the empirical Fisher cautions against treating it as a generic substitute for an arbitrary objective Hessian [Kunstner et al., 2019]. Checkpoint-based influence estimation further shows how stored checkpoints can expose training effects along the optimization trajectory [Pruthi et al., 2020]. Existing curvature-based analyses typically characterize isolated update sensitivity and can be costly to apply online at LLM scale. In contrast, we focus on the directly observable interaction between consecutive updates in task-conditioned output space.

3 Cross-Step Dynamics

We use a *token log-probability footprint* to denote the vector of log-probability changes that a parameter update induces on a fixed set of realized response-token coordinates. Each coordinate (x_i, a_i) pairs a realized token a_i with its prefix context x_i . Accordingly, task-conditioned output space refers here to the vector of token log-probabilities evaluated on such coordinates from the domain being analyzed. We now formalize the mixed output interaction between the preceding realized update and the current base update in this output space. Consider

$$\theta_{k-1} \xrightarrow{\delta_{k-1}} \theta_k \xrightarrow{\delta_k^0} \theta_{k+1}^0, \quad (1)$$

where $\delta_{k-1} := \theta_k - \theta_{k-1}$ is the preceding realized update and $\delta_k^0 := \theta_{k+1}^0 - \theta_k$ is the current base update before the OSOL correction. Focusing on the mixed output interaction between adjacent updates makes the relevant geometry checkpoint-observable, token-resolved, and amenable to online control.

3.1 Output Geometry

Let $\mathcal{E}_k = \{(x_i, a_i)\}_{i=1}^N$ denote eligible response-token coordinates from the domain selected for cross-step control at iteration k , which we call the *focus domain*; Section 4 gives its role in the training update and Section 4.1 specifies the implementation-level eligibility rule. Let $\ell_k(\theta) := [\log \pi_\theta(a_i | x_i)]_{i=1}^N$ collect the corresponding token log-probabilities. Define the output KL on these focus-domain token contexts

$$\mathcal{D}_k(\delta) := \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\pi_{\theta_k}(\cdot | x_i) \parallel \pi_{\theta_k + \delta}(\cdot | x_i)). \quad (2)$$

For $\varphi_{i,a} := \nabla \log \pi_{\theta_k}(a | x_i)$ and $\varphi_i := \varphi_{i,a_i}$, define

$$\bar{\mathbf{F}}_k := \frac{1}{N} \sum_i \mathbb{E}_{a \sim \pi_{\theta_k}(\cdot | x_i)} [\varphi_{i,a} \varphi_{i,a}^\top], \quad \mathbf{F}_k := \frac{1}{N} \sum_i \varphi_i \varphi_i^\top. \quad (3)$$

The first, which we call the output Fisher, averages score outer products over the full next-token distribution at each context. The second is the empirical output Fisher obtained from the response tokens actually realized in the rollout [Amari, 1998, Martens, 2020, Kunstner et al., 2019]. This distinction lets the KL geometry be defined over the policy distribution while keeping the cross-step interaction observable on realized tokens.

Lemma 1 (Task-conditioned output Hessian). *Under the local regularity conditions in Appendix C.1,*

$$\mathcal{D}_k(\mathbf{0}) = 0, \quad \nabla \mathcal{D}_k(\mathbf{0}) = \mathbf{0}, \quad \nabla^2 \mathcal{D}_k(\mathbf{0}) = \bar{\mathbf{F}}_k, \quad \mathcal{D}_k(\delta) = \frac{1}{2} \delta^\top \bar{\mathbf{F}}_k \delta + O(\|\delta\|^3). \quad (4)$$

Thus the relevant geometry is the exact local Hessian of output KL—the metric underlying natural-gradient and trust-region policy optimization [Kakade, 2001, Schulman et al., 2015]—rather than an identification of the complete objective Hessian with the Fisher. Proofs are in Appendices C.3 and C.4.

On the same realized coordinates, define the preceding footprint and the current base footprint

$$\mathbf{u} := \ell_k(\theta_k) - \ell_k(\theta_{k-1}), \quad \mathbf{d}^0 := \ell_k(\theta_{k+1}^0) - \ell_k(\theta_k). \quad (5)$$

Theorem 1 (Checkpoint token–Fisher duality). *If $\|\delta_{k-1}\|, \|\delta_k^0\| = O(\eta)$, then*

$$\frac{1}{N} \mathbf{u}^\top \mathbf{d}^0 = \delta_{k-1}^\top \mathbf{F}_k \delta_k^0 + O(\eta^3). \quad (6)$$

The inner product of the preceding and current base footprints therefore exposes the mixed output interaction between the preceding realized update and the current base update under the empirical output Fisher. Both footprints are defined on the same realized token coordinates, while the online controller needs only the already available preceding footprint $\mathbf{u}$. The coordinatewise expansion and explicit finite-step error are given in Appendices D.1–D.3.

Token Backtracking. The interaction has the exact path interpretation

$$\Gamma(\mathbf{u}, \mathbf{d}) := -\frac{1}{N} \mathbf{u}^\top \mathbf{d} = \frac{\|\mathbf{u}\|^2 + \|\mathbf{d}\|^2 - \|\mathbf{u} + \mathbf{d}\|^2}{2N}. \quad (7)$$

Hence $\Gamma > 0$ measures stepwise output motion that is canceled in the net displacement. Together with Theorem 1,

$$\Gamma(\mathbf{u}, \mathbf{d}^0) = -\delta_{k-1}^\top \mathbf{F}_k \delta_k^0 + O(\eta^3), \quad (8)$$

which gives the mixed output interaction a direct trajectory interpretation. Appendix D.5 decomposes the global quantity as $\Gamma = \text{Rev} - \text{Align}$, where $\text{Rev} = \mathcal{B}^- + \mathcal{B}^+$ separates token reversals from aligned motion. The nonpositive history correction used by OSOL has a monotone reduction guarantee for the negative-to-positive component; see Appendix F.1. We therefore define the controlled backtracking mass as

$$\mathcal{B}^-(\mathbf{u}, \mathbf{d}) := \frac{1}{N} \sum_{i: u_i < 0} (-u_i) [d_i]_+, \quad (9)$$

which captures negative-to-positive rebound: a token log-probability decreased by the preceding update and immediately increased by the next one.

3.2 Backtracking Control

OSOL applies a nonpositive residual on $\mathcal{C}_k := \{i \in \mathcal{E}_k : u_i < 0\}$. Let $\xi_i \leq 0$ denote its effective post-clipping change to the policy coefficient; monotonicity of scalar clipping preserves this sign (Appendix E.1).

Lemma 2 (History-conditioned PPO penalty). *For $z_i := \log(\pi_{\theta}(a_i | x_i)/\pi_{\theta_k}(a_i | x_i))$, the token-level loss difference at the PPO anchor satisfies*

$$\mathcal{J}_i^{\text{OSOL}}(z_i) - \mathcal{J}_i^0(z_i) = -e^{z_i}\xi_i, \quad \partial_{z_i}(\mathcal{J}_i^{\text{OSOL}} - \mathcal{J}_i^0)|_{z_i=0} = -\xi_i \geq 0. \quad (10)$$

Thus gradient descent adds a history-conditioned penalty against increasing the selected likelihood ratios. Appendix E.2 derives the result from the clipped PPO surrogate [Schulman et al., 2017].

Let $\Phi_{\mathcal{E}} := [\varphi_1, \dots, \varphi_N]$ and $\mathbf{K}_{\mathcal{E}} := \Phi_{\mathcal{E}}^{\top} \Phi_{\mathcal{E}}$, where $\mathbf{K}_{\mathcal{E}}$ is the token-gradient similarity matrix and captures cross-token coupling induced by shared model parameters. In the local policy-gradient model, the correction induces

$$\mathbf{h} := \Phi_{\mathcal{E}}^{\top} \delta_k^r = \frac{\eta}{|\mathcal{R}|} \mathbf{K}_{\mathcal{E}} \xi, \quad (11)$$

where $\mathbf{h}$ is the correction footprint.

Because all tokens share model parameters, a negative coefficient correction does not by itself guarantee a negative log-probability change for every selected token: cross-token interactions in $\mathbf{K}_{\mathcal{E}}$ can alter the local response. The following sufficient condition requires each selected token’s own local response to dominate the aggregate cross-token effect.

Lemma 3 (Residual-to-output sign transfer). *Under the local dominance condition in Appendix E.4, $h_i \leq 0$ for the selected coordinates.*

Under this condition, the nonpositive coefficient correction transfers to a nonpositive output-space correction on the selected tokens. Appendix E.4 gives the exact condition and its extension to the locally preconditioned optimizer response.

Proposition 1 (Backtracking contraction). *If $h_i \leq 0$ on $\mathcal{C}_k$, then*

$$\Delta_k^{\text{ctrl}} := \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0) - \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0 + \mathbf{h}) \geq 0. \quad (12)$$

For the corrected footprint $\mathbf{d}^{\text{OSOL}}$ with $\|\delta_k^r\| = O(\eta)$,

$$\mathcal{B}^-(\mathbf{u}, \mathbf{d}^{\text{OSOL}}) \leq \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0) - \Delta_k^{\text{ctrl}} + O(\eta^3). \quad (13)$$

The proof and explicit finite-step remainder are in Appendices E.5 and E.6. The theory therefore closes an observable-to-controllable loop: output KL identifies the local geometry, the preceding and current base footprints expose the mixed output interaction, token signs localize backtracking, and the history-conditioned penalty contracts the negative-to-positive component targeted by OSOL.

4 Method Design

One Step, One Lead. Each mixed-domain update designates exactly one domain f_k as the *focus domain*; the remaining domains are *non-focus domains*. The focus domain receives a larger task weight in the base GRPO coefficient, and only its eligible response tokens receive the history-based correction. The non-focus domains remain in the same update with smaller task weights, so the method prioritizes one domain without discarding the others. A training schedule selects the focus domain at each iteration.

At iteration k , OSOL samples the standard mixed-domain rollout batch from π_{θ_k} and rescores the same realized eligible focus-domain tokens with the frozen preceding checkpoint θ_{k-1} . The resulting drift defines the intervention set ($u_i < 0$), its rank determines relative strength, and τ sets the batchwise scale. The residual is then added to the existing GRPO coefficient and optimized with the unchanged clipped policy objective. Thus OSOL modifies the update signal rather than introducing a separate auxiliary loss. All history-dependent quantities are stop-gradient. The focus assignment determines where the checkpoint controller is evaluated; it does not alter the definition of the underlying group-relative reward signal.

4.1 Update History

Let A_i^g be the GRPO group-normalized trajectory advantage broadcast to token i , and let $\text{src}(i)$ denote its domain. With focus and non-focus weights $w_f > w_{nf}$, the task-weighted base coefficient is

$$A_i^{\text{base}} = c_i A_i^g, \quad c_i = \frac{w_{\text{src}(i)}}{Z_k}, \quad Z_k = \frac{w_f N_k^f + w_{nf} N_k^{nf}}{N_k^f + N_k^{nf}}, \quad (14)$$

where N_k^f and N_k^{nf} are focus- and non-focus-token counts. The normalization keeps the coefficient scale comparable as the designated focus changes. The history residual is added after this weighting and is not multiplied by c_i again.

Let $\mathcal{E}_k$ contain current focus-domain response tokens with nonzero trajectory-level GRPO advantage. On each realized coordinate (x_i, a_i) , compute

$$u_i = \log \pi_{\theta_k}(a_i | x_i) - \log \pi_{\theta_{k-1}}(a_i | x_i), \quad \mathcal{C}_k = \{i \in \mathcal{E}_k : u_i < 0\}. \quad (15)$$

Thus u is the preceding footprint on the current rollout coordinates, not a prediction of the next update. Both checkpoints score the same (x_i, a_i) pairs, so the comparison requires no token matching or auxiliary model. A token belongs to $\mathcal{C}_k$ when the previous mixed-domain step reduced its current realized log-probability. This defines the negative-drift candidate set on which a nonpositive correction monotonically suppresses negative-to-positive rebound (Appendix F.1).

4.2 Residual Construction

For $i \in \mathcal{C}_k$, let $\text{rank}_0(u_i)$ be the ascending zero-based rank and set

$$p_i = \frac{\text{rank}_0(u_i) + \frac{1}{2}}{|\mathcal{C}_k|}, \quad \bar{r}_i = r_{\min} + \frac{\text{clip}(p_i, p_{\text{lo}}, p_{\text{hi}}) - p_{\text{lo}}}{p_{\text{hi}} - p_{\text{lo}}} (r_{\max} - r_{\min}), \quad (16)$$

with $\bar{r}_i = 0$ off $\mathcal{C}_k$. We use $(p_{\text{lo}}, p_{\text{hi}}, r_{\min}, r_{\max}) = (0.1, 1, -1, 0)$, assigning stronger penalties to more negative drift. This ordering is locally optimal for allocating a fixed collection of penalty levels to active rebound coordinates (Appendix F.2). Using ranks rather than raw drift magnitudes also reduces sensitivity to batch-to-batch scale changes and isolated extreme log-probability shifts.

Define the supported profile and its adaptive scale by

$$q_i := \bar{r}_i \mathbf{1}\{i \in \mathcal{C}_k\}, \quad (17)$$

$$\kappa_k := \tau \frac{\text{std}_{\mathcal{E}_k}(A^{\text{base}})}{\text{std}_{\mathcal{E}_k}(q)}, \quad r_i^{\text{OSOL}} := \kappa_k q_i. \quad (18)$$

Whenever the statistics are nondegenerate,

$$\text{std}_{\mathcal{E}_k}(r^{\text{OSOL}}) = \tau \text{std}_{\mathcal{E}_k}(A^{\text{base}}). \quad (19)$$

The standard setting uses $\tau = 0.03$. Hence τ controls the residual relative to the current base-coefficient scale rather than to the raw scale of the preceding drift. Numerical fallback and the code-level gate and mask are specified in Appendix F.3–F.4.

The final policy coefficient is

$$\tilde{A}_i = \text{clip}(A_i^{\text{base}} + r_i^{\text{OSOL}}, -3, 3). \quad (20)$$

With $\omega_i(\theta) := \pi_{\theta}(a_i | x_i) / \pi_{\theta_{\text{old}}}(a_i | x_i)$, it is optimized by the unchanged clipped surrogate [Schulman et al., 2017]:

$$\mathcal{L}_{\text{OSOL}}(\theta) = -\mathbb{E} \left[\frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} \min \left(\omega_i(\theta) \tilde{A}_i, \text{clip}(\omega_i(\theta), 1 - \epsilon_c, 1 + \epsilon_c) \tilde{A}_i \right) \right]. \quad (21)$$

Thus, OSOL converts the preceding token drift into a tokenwise history-conditioned correction within the standard GRPO objective.

5 Experiments

5.1 Setup

Models and baselines. We evaluate OSOL on Qwen3-30B-A3B and Qwen3-8B-Base. On both backbones, we compare against the pre-RL base model, standard GRPO [Guo et al., 2025], GRPO+KL_{0.001} as a conservative-update baseline, MGS [Cai et al., 2026] as a first-order module-level gradient-surgery baseline, and CGPO [Liang et al., 2026] as a curvature-guided multi-domain optimization baseline. All methods follow the same evaluation protocol. Complete main-training and generation settings are summarized in Appendix F.5.

Training data. We construct a training dataset mixture covering four domains: mathematics, code generation, instruction following (IF), and open-ended chat (Chat). The mathematics and code examples are quality-filtered subsets of the NVIDIA Nemotron Post-Training Dataset v2 [Basant et al., 2025]. For Chat, we use the WildChat-IF subset provided by MGS, which is derived from the Tulu 3 WildChat-IF dataset. For IF, we use AllenAI’s RLVR-IFeval, whose prompts contain programmatically verifiable constraints. Each domain contributes 7.5K training examples, resulting in a balanced mixture across all four domains.

Evaluation. We evaluate mathematical reasoning on AIME 2024, AIME 2025, and AMC 2023; code generation on LiveCodeBench; IF on IFBench; and Chat on WildBench-Hard and CreativeWriting v3. For AIME, AMC, and LiveCodeBench, we report avg@8. WildBench-Hard and CreativeWriting v3 are reported on their normalized scales. Because the training mixture contains four equally represented domains, we use a domain-macro average as the primary aggregate metric. We first average benchmarks within mathematics and Chat, then assign equal weight to the four domain scores. This prevents a domain from receiving greater aggregate weight solely because it is represented by more benchmarks.

5.2 Main Results

Table 1: Main results on the Qwen3-30B-A3B mixture-of-experts backbone. Math averages AIME24, AIME25, and AMC23 avg@8; Code uses LiveCodeBench avg@8; IF uses IFBench strict-prompt accuracy; Chat averages normalized WildBench-Hard and CreativeWriting v3 scores. Domain Avg. is the unweighted mean of the four domain scores. Higher is better. Best and second-best values are bolded and underlined.

Method	Math	Code	IF	Chat	Domain Avg.
<i>Base model</i>	0.5312	0.3310	0.2100	0.6392	0.4279
GRPO	0.5632	0.2603	0.2066	0.6900	0.4300
GRPO+KL _{0.001}	0.5538	0.2540	0.1900	0.7271	0.4312
MGS	<u>0.5757</u>	<u>0.3858</u>	0.2000	0.6625	<u>0.4560</u>
CGPO	0.5670	0.3835	<u>0.2067</u>	0.6541	0.4528
OSOL (ours)	0.5822	0.4529	0.1933	<u>0.7003</u>	0.4822

Mixture-of-experts backbone. Table 1 shows that MGS and CGPO achieve domain-macro averages of 0.4560 and 0.4528 on Qwen3-30B-A3B, respectively, substantially exceeding standard GRPO at 0.4300. OSOL further reaches 0.4822, improving over MGS, the strongest baseline, by 0.0262 points or 5.7%. OSOL attains the best scores in mathematics and code and ranks second in Chat, while its IF score remains below the pre-RL base model. These results demonstrate that cross-step control improves over both first-order gradient surgery and curvature-guided optimization on a sparsely activated mixture-of-experts backbone.

Dense backbone. On Qwen3-8B-Base, Table 2 shows that OSOL achieves the highest domain-macro average of 0.3735. This corresponds to an improvement of 0.0224 points (6.4%) over standard GRPO and 0.0035 points (0.9%) over GRPO+KL_{0.001}, the strongest compared baseline. OSOL achieves the highest mathematics score and ranks

Table 2: Main results on Qwen3-8B-Base. Domain definitions and averaging follow Table 1. Higher is better. Best and second-best values are bolded and underlined.

Method	Math	Code	IF	Chat	Domain Avg.
<i>Base model</i>	0.1316	0.1751	0.1433	0.4466	0.2241
GRPO	0.2906	0.2119	0.2733	0.6284	0.3511
GRPO+KL _{0.001}	0.2750	0.2249	0.3000	0.6802	<u>0.3700</u>
MGS	0.2948	0.0883	0.2833	0.6401	0.3266
CGPO	<u>0.3094</u>	0.1828	0.2567	0.6623	0.3528
OSOL (ours)	0.3111	<u>0.2173</u>	<u>0.2933</u>	<u>0.6723</u>	0.3735

second in code, IF, and Chat. GRPO+KL_{0.001} remains slightly stronger in these three domains, while its lower mathematics score results in a smaller equal-domain average. The aggregate result therefore reflects a favorable cross-domain trade-off despite the absence of uniform per-domain dominance.

Table 3: Ablation results on Qwen3-8B-Base. OSOL without history residual retains the one-step-one-lead focus structure while removing the checkpoint-history correction; the remaining variants ablate focus weighting, drift-based ranking, or token-level allocation.

Method	Math	Code	IF	Chat	Domain Avg.
OSOL without history residual	0.2996	<u>0.2114</u>	<u>0.2966</u>	0.6324	0.3600
Equal task weights	0.0615	0.0202	0.2467	0.3978	0.1815
Response-level residual	0.3202	0.2021	0.3000	<u>0.6338</u>	<u>0.3640</u>
Random residual	0.0698	0.0390	0.2400	<u>0.3866</u>	<u>0.1838</u>
OSOL (ours)	<u>0.3111</u>	0.2173	0.2933	0.6723	0.3735

5.3 Ablation Analysis

The matched OSOL without history residual variant retains the same one-step-one-lead focus schedule and asymmetric focus-non-focus base weighting while setting the checkpoint-history residual to zero. It reaches a domain-macro average of 0.3600, above standard GRPO at 0.3511 (Table 2) but 0.0135 below the full method at 0.3735. This comparison clarifies the role of the one-step-one-lead structure in the full design: focus scheduling and asymmetric weighting establish a structured cross-step trajectory with a designated focus domain, while the checkpoint-history residual uses the preceding update to counter sequential rebound. The full method improves over this structure-only baseline by 3.8%, indicating that focus scheduling alone does not account for the observed gain.

The remaining ablations isolate how this cross-step correction is implemented. Equal task weights keeps the focus-restricted residual but removes asymmetric base weighting, reducing the domain-macro average to 0.1815 (−51.4%); Random residual keeps the correction pathway but replaces drift-based token ranking with random weights, yielding 0.1838 (−50.8%). The large degradations indicate that the history correction depends on both the focus-domain weighting and the preceding footprint used to localize the correction. Response-level residual retains both the one-step-one-lead structure and checkpoint signal but removes token-specific allocation within each response, reaching 0.3640. The full token-level method further improves this score by 0.0095 points (2.6%), mainly through gains in code and Chat. Together, the ablations support the intended decomposition of OSOL: one-step-one-lead organizes the cross-step trajectory, checkpoint drift localizes rebound risk, and token-level allocation applies the correction at the granularity where that risk is observed.

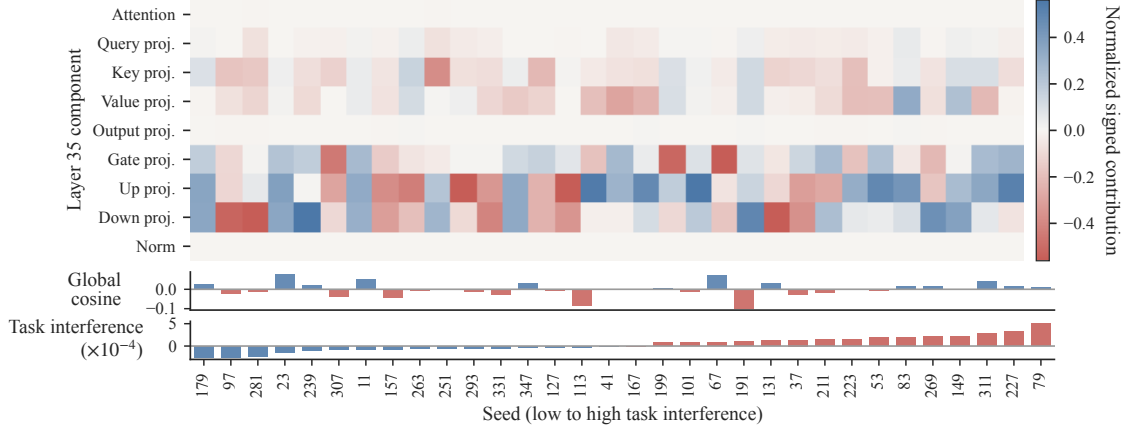


Figure 3: **Layer-35 signed gradient contributions expose cancellation hidden by the global cosine.** Columns are seeds ordered by observed task interference; opposing module contributions can cancel under global aggregation.

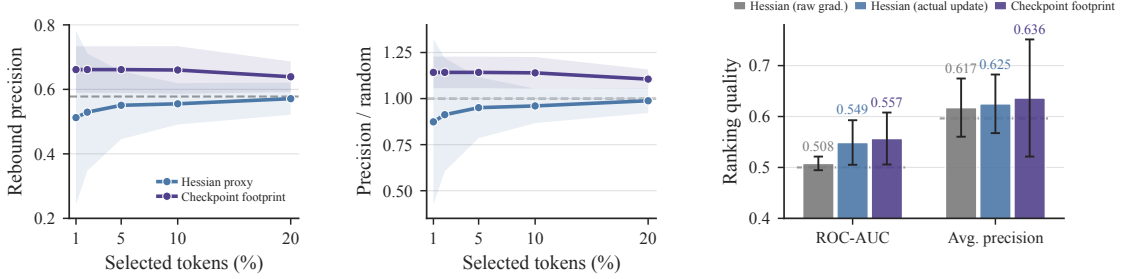


Figure 4: Checkpoint history provides a stronger token-level rebound signal than Hessian-based proxies. Checkpoint-footprint ranking improves precision and enrichment across candidate fractions (left) and remains superior in ROC-AUC and average precision even when the Hessian proxy uses the realized optimizer update (right).

5.4 Mechanism Analysis

Section 3 characterizes cross-step backtracking using two consecutive token footprints, whereas the online controller must act before the current update is completed and therefore observes only the preceding footprint u . We therefore evaluate whether u alone contains sufficient information to identify tokens likely to rebound under the next update. As shown in Figure 5, within the negative-drift candidate set, checkpoint-footprint ranking achieves 0.567 ROC-AUC and 0.627 average precision, compared with 0.498 and 0.579 for the Hessian proxy, while improving top-5% precision from 0.551 to 0.661; the non-token-resolved global reference yields the expected ROC-AUC of 0.500. These results indicate that preceding negative drift provides an informative token-level ranking of future rebound risk. Figure 3 further explains why this information can be obscured by global gradient aggregation: large opposing module-level contributions, particularly in MLP projections, cancel to produce near-zero global cosine values despite substantially different levels of task interference. Figure 4 further evaluates the robustness of the token-level ranking. Checkpoint-footprint ranking remains stronger across selected candidate fractions from 1% to 20%, with the largest advantage among the highest-ranked candidates, supporting the drift-ranked token allocation in Eq. (16). On the three-seed control subset, replacing the raw gradient with the realized optimizer displacement improves the Hessian proxy’s ROC-AUC from 0.508 to 0.549, but checkpoint ranking remains higher at 0.557. Together, these results show that the preceding checkpoint footprint provides a direct pre-update signal for identifying rebound-prone tokens and allocating OSOL’s correction strength without explicit Hessian estimation.

6 Conclusion

Multi-domain RL can suffer sequential interference even when same-point gradients appear weakly conflicting. We introduced a cross-step view in which adjacent checkpoint footprints expose the mixed output-space interaction between consecutive updates and give harmful reversals a direct interpretation as cross-step output backtracking. Building on this view, OSOL designates a focus domain and adds a drift-ranked, scale-normalized token-level history residual to the standard GRPO coefficient, requiring neither auxiliary supervision nor explicit higher-order differentiation. Our analysis characterizes when this correction contracts the targeted backtracking component. Controlled studies show that cross-step backtracking is more strongly associated with subsequent task damage than same-point gradient diagnostics, while the preceding checkpoint footprint identifies future token rebound more effectively than Hessian-based proxies. On Qwen3-30B-A3B and Qwen3-8B-Base, OSOL achieves domain-macro averages of 0.4822 and 0.3735, with relative gains of 5.7% and 0.9% over the strongest compared baselines, respectively. Ablations further verify the importance of focus–non-focus base weighting, drift-guided localization, and token-level allocation. Together, these results support cross-step output-space regulation as a practical first-order approach to mitigating interference in multi-domain policy optimization.

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A Controlled Diagnostic Study Protocols

This section provides the complete protocols for the two controlled studies used to motivate and analyze OSOL. Both studies instantiate adjacent updates explicitly, preserve the optimizer state across the two steps, and evaluate every diagnostic against quantities measured from the realized checkpoint trajectory. The first study evaluates which replicate-level signal best explains downstream cross-domain interference. The second evaluates which token-level signal best identifies tokens whose preceding output movement is reversed by the next update.

A.1 Common model, data, and one-step optimization

Model and data. We use Qwen3-4B as the policy in both studies. Math prompts are sampled from the math portion of the same filtered NVIDIA Nemotron Post-Training Dataset v2 mixture used for the main experiments, and chat prompts are sampled from the MGS-provided WildChat-IF subset. Math generations are scored by the task verifier and chat generations are scored by Skywork-Reward-Llama-3.1-8B-v0.2. For each prompt, we sample four responses with temperature 1.0 and nucleus threshold $p = 0.95$. Prompts are truncated to at most 1,024 tokens and responses to at most 512 tokens. We disable explicit thinking-mode prompting so that all diagnostics are computed under the same response format.

Advantages and updates. Within each four-response group, rewards are standardized to produce the GRPO trajectory advantage. If every response in a group receives the same verifier reward, we use a unit coefficient for that group, yielding a deterministic nonzero one-step update while preserving the sampled response distribution. Chat groups use reward-model scores and have nonzero within-group variation in the analyzed runs. Each replicate performs two consecutive AdamW steps with learning rate 10^{-6} , weight decay 0.1, $(\beta_1, \beta_2) = (0.9, 0.999)$, numerical constant 10^{-8} , and gradient-norm clipping at 1.0. Computation uses bfloat16, and log probabilities are accumulated over response tokens with padding coordinates masked as appropriate.

Controlled output-facing subspace. The repeated gradient, optimizer, and finite-difference evaluations are carried out in an output-facing parameter subspace consisting of the final transformer block and the language-model head. This subspace contains 100,930,816 parameters, or 2.51% of Qwen3-4B, and is shared by the compared diagnostics in the primary studies. The coverage control in Appendix A.4 expands this subspace to the final four and final eight transformer blocks while retaining the language-model head. This design permits exact replicate-level replay of gradients, optimizer displacements, and token log probabilities while keeping the policy, sampled trajectories, rewards, and evaluation targets matched.

A.2 Study I: same-point diagnostics and realized task interference

Replicate construction. We construct 32 independent replicates from a common initial checkpoint θ_0 . Randomness across replicates comes from prompt selection and response sampling; the initial policy and all optimization hyperparameters are fixed. Each replicate uses two math update prompts and two chat update prompts, with four responses per prompt. In addition, it samples two math prompts for the token-motion probe and two disjoint math prompts for the damage probe. The update, motion-probe, and damage-probe prompt sets are mutually disjoint within a replicate.

Starting from θ_0 , we first apply the math update and then the chat update:

$$\theta_0 \xrightarrow{\text{math update}} \theta_1 \xrightarrow{\text{chat update}} \theta_2. \quad (22)$$

Math responses and both math probe sets are sampled at θ_0 . After the math step, chat responses are sampled from θ_1 , scored by the reward model, and used for the $\theta_1 \rightarrow \theta_2$ update. To obtain a same-point gradient comparison without changing the realized chat trajectories, the fixed θ_1 chat responses, rewards, and advantages are replayed at θ_0 when computing the chat gradient. The same AdamW optimizer instance is retained across the math and chat steps, matching the stateful sequential-update dynamics.

Independent interference target. Let $\mathcal{P}_{\text{dmg}}$ denote the held-out math damage probe. We measure the effect of the chat update by the change in fixed-response token negative log likelihood:

$$\text{Damage}_{\text{math} \leftarrow \text{chat}} := \mathcal{L}_{\text{NLL}}(\theta_2; \mathcal{P}_{\text{dmg}}) - \mathcal{L}_{\text{NLL}}(\theta_1; \mathcal{P}_{\text{dmg}}). \quad (23)$$

A positive value indicates that the chat step increases math-probe NLL. Because $\mathcal{P}_{\text{dmg}}$ is disjoint from the update and motion probes, the dependent variable is not reused in any diagnostic score.

Same-point global and module diagnostics. Using the fixed update batches, define the descent directions at θ_0 as

$$\mathbf{v}_m := -\nabla \mathcal{L}_m(\theta_0), \quad \mathbf{v}_c := -\nabla \mathcal{L}_c(\theta_0). \quad (24)$$

The global first-order diagnostic is

$$s_{\text{global}} = \left[-\frac{\langle \mathbf{v}_m, \mathbf{v}_c \rangle}{\|\mathbf{v}_m\|_2 \|\mathbf{v}_c\|_2} \right]_+. \quad (25)$$

For the module diagnostic, partition the output-facing subspace into attention projections, MLP projections, normalization parameters, and the language-model head. If $\mathbf{v}_m^{(g)}$ and $\mathbf{v}_c^{(g)}$ are the directions in group g , define

$$s_{\text{module}} = \frac{\sum_g [-\langle \mathbf{v}_m^{(g)}, \mathbf{v}_c^{(g)} \rangle]_+}{\sum_g |\langle \mathbf{v}_m^{(g)}, \mathbf{v}_c^{(g)} \rangle|}. \quad (26)$$

This normalized negative dot-product mass retains opposing module contributions that cancel in the aggregate inner product.

Cross-step backtracking diagnostic. On the disjoint math motion probe $\mathcal{P}_{\text{mot}}$, we evaluate the same realized response tokens at all three checkpoints. For token i , let

$$u_i = \log \pi_{\theta_1}(a_i | x_i) - \log \pi_{\theta_0}(a_i | x_i), \quad d_i = \log \pi_{\theta_2}(a_i | x_i) - \log \pi_{\theta_1}(a_i | x_i). \quad (27)$$

The backtracking mass used in Figure 6 is

$$s_{\text{rev}} = \frac{1}{|\mathcal{R}_{\text{mot}}|} \sum_{i \in \mathcal{R}_{\text{mot}}} \mathbf{1}\{u_i < 0\} [-u_i d_i]_+ = \frac{1}{|\mathcal{R}_{\text{mot}}|} \sum_{i \in \mathcal{R}_{\text{mot}}} (-u_i) [d_i]_+ \mathbf{1}\{u_i < 0\}, \quad (28)$$

where $\mathcal{R}_{\text{mot}}$ contains response-token coordinates. The diagnostic therefore records the mass of positive rebound following a negative preceding movement, using the realized optimizer steps rather than a virtual parameter perturbation.

Statistical analysis. Each replicate contributes one value of $\text{Damage}_{\text{math} \leftarrow \text{chat}}$ and one value for each diagnostic. We use Spearman rank correlation as the primary association statistic and display a Theil–Sen trend in the scatter plots. Confidence intervals are obtained with 50,000 paired replicate bootstrap resamples. The same resampled replicate indices are used when comparing two diagnostics, so the reported correlation gains are paired differences rather than differences between independently estimated intervals. We additionally recompute each correlation after removing every replicate once. The resulting leave-one-replicate-out ranges are reported in Figure 8.

Observed diagnostic ordering. The global and module diagnostics have correlations of -0.19 and -0.22 with downstream interference, whereas backtracking mass reaches $\rho_s = 0.40$ with a 95% bootstrap interval of $[0.05, 0.69]$. Its paired correlation gains are $+0.59$ over the global diagnostic and $+0.63$ over the module diagnostic. The sign and ordering are preserved under every leave-one-replicate-out recomputation. Figure 3 further decomposes the same replicates and shows how positive and negative module contributions disappear under global aggregation.

A.3 Study II: token-level rebound identification

Causal triplets and token table. The second study uses eight independent seeds and the same Qwen3-4B one-step configuration. Each seed samples two math prompts and two chat prompts with four responses per prompt. The math step produces θ_1 from θ_0 . Chat responses are then sampled from θ_1 , scored by the reward model, and used both as the token probe and as the update batch that produces θ_2 . Therefore, u_i and d_i in Eq. (27) are evaluated on the exact token coordinates responsible for the second update.

We retain chat response tokens with nonzero trajectory-level advantage:

$$\mathcal{E} = \{i : i \text{ is a response token, } A_i \neq 0\}. \quad (29)$$

The checkpoint-history candidate set and realized rebound label are

$$\mathcal{C} = \{i \in \mathcal{E} : u_i < 0\}, \quad y_i = \mathbf{1}\{u_i < 0, d_i > 0\}, \quad q_i = (-u_i)[d_i]_+. \quad (30)$$

The paper-facing comparison is conditioned on $\mathcal{C}$: it asks which negative-drift tokens are most likely to rebound under the subsequent unregulated update. Across the eight seeds, 22.6% of eligible tokens enter $\mathcal{C}$, and the mean rebound prevalence within $\mathcal{C}$ is 57.8%.

Checkpoint-footprint score. Within $\mathcal{C}$, the raw checkpoint score is $[-u_i]_+$. The rank-based score used by OSOL sorts candidate drifts in ascending order and assigns

$$p_i = \frac{\text{rank}_0(u_i) + 1/2}{|\mathcal{C}|}, \quad s_i^{\text{ckpt}} = 1 - \frac{\text{clip}(p_i, 0.1, 1) - 0.1}{0.9}, \quad (31)$$

with score zero outside $\mathcal{C}$. More negative drift receives a larger score, while the strongest decile shares the maximum score. This is the same ordering and quantile profile used by the online controller before batchwise scale normalization.

Finite-difference Hessian proxy. We compare checkpoint history with a token-level curvature proxy constructed on the same fixed chat trajectories, rewards, and advantages. Let $\mathcal{L}_c^{\text{loc}}$ be the unclipped local policy-gradient surrogate on this fixed chat batch, and let

$$\mathbf{v}_m := -\nabla \mathcal{L}_m(\boldsymbol{\theta}_0), \quad \mathbf{v}_c := -\nabla \mathcal{L}_c^{\text{loc}}(\boldsymbol{\theta}_0). \quad (32)$$

For selected parameters $\boldsymbol{\theta}_S$, choose

$$\alpha = \epsilon \frac{\|\boldsymbol{\theta}_S\|_2}{\|\mathbf{v}_m\|_2} \quad (33)$$

and estimate the directional Hessian-vector product by the centered difference

$$\mathbf{h} \approx \frac{\nabla \mathcal{L}_c^{\text{loc}}(\boldsymbol{\theta}_0 + \alpha \mathbf{v}_m) - \nabla \mathcal{L}_c^{\text{loc}}(\boldsymbol{\theta}_0 - \alpha \mathbf{v}_m)}{2\alpha}. \quad (34)$$

The main sweep uses $\epsilon \in \{10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}\}$; the primary score uses the pre-specified summary entry at $\epsilon = 10^{-4}$. Define

$$\boldsymbol{\theta}_m = \boldsymbol{\theta}_0 + \alpha \mathbf{v}_m, \quad \boldsymbol{\theta}_{\text{pred}} = \boldsymbol{\theta}_m + \alpha(\mathbf{v}_c - \alpha \mathbf{h}). \quad (35)$$

For probe token i , write $\ell_i(\boldsymbol{\theta}) := \log \pi_{\boldsymbol{\theta}}(a_i | x_i)$. The virtual math displacement and curvature-adjusted predicted chat displacement are

$$\Delta_i^m = \ell_i(\boldsymbol{\theta}_m) - \ell_i(\boldsymbol{\theta}_0), \quad \Delta_i^{\text{pred}} = \ell_i(\boldsymbol{\theta}_{\text{pred}}) - \ell_i(\boldsymbol{\theta}_m), \quad (36)$$

and the Hessian ranking score is

$$s_i^H = [-\Delta_i^m \Delta_i^{\text{pred}}]_+. \quad (37)$$

All virtual log probabilities are computed on the same tokenized probe batch. We also record the induced forward policy KL for each virtual displacement, providing an output-space check on the numerical perturbation scale.

Ranking metrics and uncertainty. For every seed, we compute receiver operating characteristic area under the curve (ROC-AUC), average precision, and precision among the top-ranked 1%, 2%, 5%, 10%, and 20% of candidates. Enrichment is precision divided by the within-candidate rebound prevalence. Average precision is evaluated at unique score thresholds, and when a top- k boundary intersects tied scores, we report expected precision under uniform tie breaking. We first compute each metric independently for each seed and then report the seed mean with a two-sided 95% Student- t interval. Paired gains use the per-seed difference between checkpoint-footprint and Hessian scores before aggregation.

Primary comparison. Within negative-drift candidates, checkpoint-footprint ranking obtains 0.567 ROC-AUC, 0.627 average precision, and 0.661 top-5% precision, compared with 0.498, 0.579, and 0.551 for the Hessian proxy. The paired gains in ROC-AUC and average precision are 0.069 and 0.048, respectively. Across selected candidate fractions, checkpoint ranking remains above the Hessian proxy in both precision and enrichment, with the clearest separation when only a small fraction of candidates is selected by OSOL.

A.4 Hessian controls and computational measurements

Realized-update direction. To separate curvature approximation from direction mismatch, we repeat the Hessian construction on seeds $\{11, 23, 37\}$ after replacing $\mathbf{v}_m$ with the selected-parameter optimizer displacement $\boldsymbol{\theta}_1 - \boldsymbol{\theta}_0$. The finite-difference procedure and token target remain unchanged. This increases Hessian ROC-AUC from 0.508 to 0.549 and average precision from 0.617 to 0.625; checkpoint footprint remains highest at 0.557 and 0.636, respectively.

Parameter-coverage sweep. We repeat the curvature proxy with the language-model head plus the final one, four, or eight transformer blocks, corresponding to 2.51%, 10.04%, and 20.07% of model parameters. Each configuration uses the same three seeds and finite-difference scales $\{3 \times 10^{-5}, 10^{-4}\}$. Peak GPU memory is measured by the maximum CUDA allocation during the complete diagnostic run. Mean peak memory rises from 26.6 to 33.8 and 43.5 GB, while Hessian ROC-AUC changes from 0.508 to 0.511 and 0.534. Checkpoint-footprint ROC-AUC on the matching trajectories is 0.557, 0.577, and 0.583.

Finite-difference scale sweep. Finally, we retain all eight primary seeds and recompute the finite-difference diagnostic for $\epsilon \in \{10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}\}$. For each scale we record the HVP norm and the forward policy KL induced by the virtual math step, the displacement induced by the negative HVP term, and predicted next step. These measurements produce Figure 9 and quantify how the numerical probe scale changes the output-space displacement being evaluated.

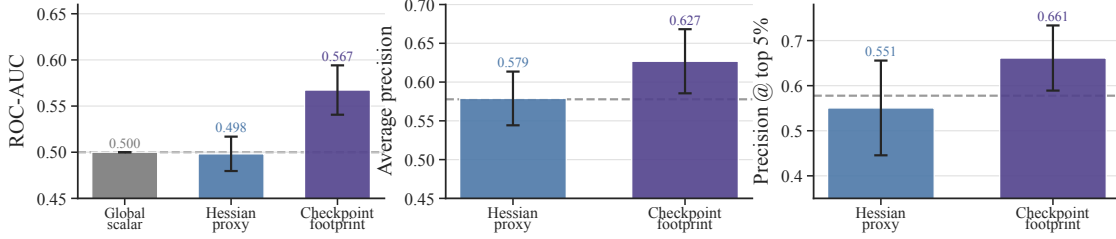


Figure 5: Rebound identification within negative-drift candidates. The checkpoint footprint outperforms the Hessian proxy in ROC-AUC, average precision, and top-5% precision; the non-token-resolved global scalar is included as a 0.5 ROC-AUC reference. Error bars show 95% confidence intervals.

B Additional Diagnostic Evidence

This section expands the controlled analyses in Appendix A and clarifies what each diagnostic reveals about cross-step interference. Study I evaluates *realized* two-step backtracking after both updates have occurred and asks whether it is associated with downstream task damage; this is a post-update diagnostic question rather than the pre-update token-ranking problem used by the online controller. Study II examines the numerical behavior of the finite-difference Hessian proxy used as a token-level comparator. Together, the analyses separate three conceptually distinct questions: whether same-point gradients order realized damage, whether the observed diagnostic ordering is stable across replicates, and whether explicit curvature estimation provides a numerically stable route to token-level interaction information.

Cross-step backtracking is more closely associated with realized task damage. Figure 6 compares the three replicate-level diagnostics from Study I against the held-out math damage measured after the subsequent chat update. The same-point global conflict score has Spearman correlation $\rho_s = -0.19$ with damage, while the module-level conflict mass gives $\rho_s = -0.22$. These negative estimates should not be interpreted as evidence that gradient conflict is beneficial; rather, within this controlled setting, neither same-point score provides a useful positive ordering of which subsequent updates damage the held-out math probe more strongly. This remains true even for the module diagnostic, which preserves opposing module contributions that disappear in the global inner product. By contrast, the realized cross-step backtracking mass reaches $\rho_s = 0.40$ with a 95% bootstrap CI of $[0.05, 0.69]$: replicates in which the second update more strongly reverses the output movement of the first tend to exhibit larger downstream damage. Importantly, the backtracking score is computed on the output movement, whereas damage is measured on a disjoint held-out probe (Appendix A.2); the association therefore does not arise from evaluating the same token set twice. The result supports the central diagnostic claim of the paper: the interaction between two *realized* updates in output space is more strongly associated with sequential task damage than same-checkpoint gradient geometry alone.

Paired uncertainty favors the trajectory-based diagnostic. Figure 7 makes the uncertainty comparison explicit. The left panel reports the bootstrap intervals of the three Spearman correlations: the intervals for the global and module diagnostics include zero, while the interval for backtracking mass lies on the positive side. More importantly, the right panel compares the diagnostics *pairwise* using identical bootstrap resamples of the 32 replicates. Backtracking mass improves correlation by $+0.59$ over the global score (95% CI $[0.05, 1.06]$) and by $+0.63$ over the module score (95% CI $[0.03, 1.15]$). Because these are paired differences, the comparison controls for replicate-to-replicate variation rather than contrasting two independently estimated confidence intervals. Combined with the module decomposition in Figure 3, this result also separates two phenomena: module-level decomposition can reveal cancellation hidden by a global cosine, but retaining that cancellation information alone is still insufficient to match the realized two-step backtracking signal in explaining downstream damage. The added value therefore comes from observing the trajectory interaction itself, not merely from computing a more granular same-point gradient statistic.

The diagnostic ordering is not driven by a single replicate. The controlled study contains 32 replicates, so we additionally assess sensitivity to individual runs rather than relying on the aggregate correlation alone. Figure 8 recomputes each Spearman correlation after removing one replicate at a time. Across all 32 leave-one-out subsets,

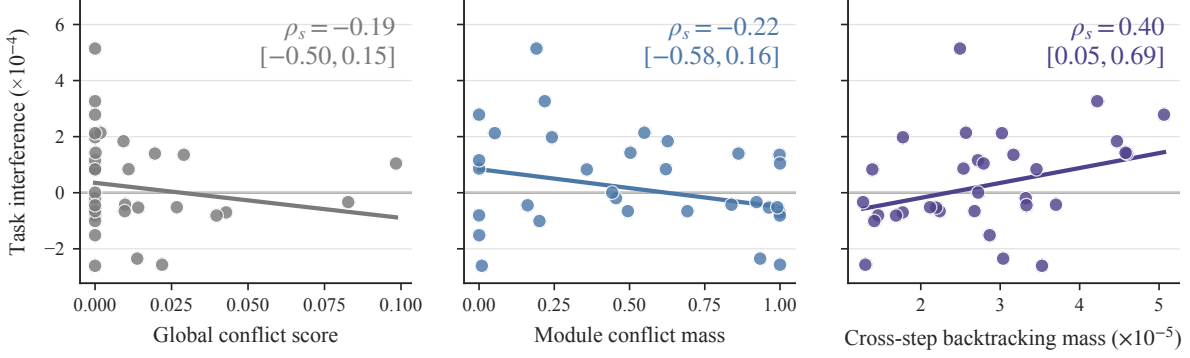


Figure 6: Replicate-level association between interference diagnostics and realized task damage. Same-point global and module scores do not positively order the subsequent damage, whereas cross-step backtracking mass shows a positive association ($\rho_s = 0.40$, 95% bootstrap CI $[0.05, 0.69]$). The backtracking and damage measurements use disjoint probes.

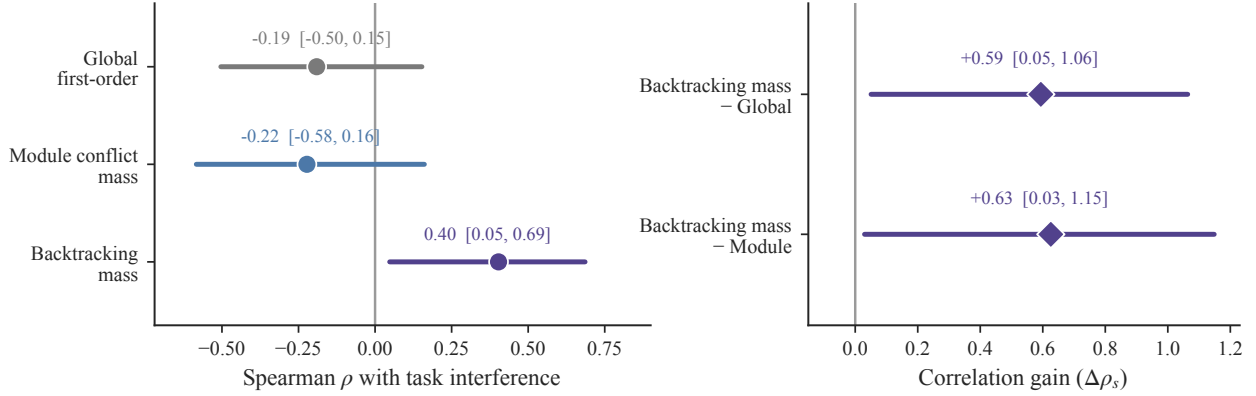


Figure 7: Uncertainty and paired correlation gains for the three Study I diagnostics. Left: bootstrap estimates of Spearman correlation with task damage. Right: paired gains from replacing the same-point global or module diagnostic with cross-step backtracking mass. Both paired intervals remain positive.

the global score remains negative in $[-0.26, -0.15]$, the module conflict mass remains negative in $[-0.31, -0.16]$, and backtracking mass remains positive in $[0.35, 0.49]$. Thus every deletion preserves both the sign pattern and the ordering of the three diagnostics. This analysis is not a substitute for an independent replication study, but it rules out the simpler explanation that the positive backtracking correlation, or the gap to the same-point baselines, is caused by one unusually influential replicate.

Finite-difference curvature introduces a consequential probe-scale choice. Figure 9 examines the finite-difference scale ϵ used to construct the Hessian-vector-product proxy in Study II. The left panel shows that the estimated HVP norm changes markedly at the smallest probe scale and then becomes flatter as ϵ increases. The right panel shows the corresponding output-space effect: the policy KL induced by the virtual focus step, the Hessian reaction, and the resulting predicted next step all increase by orders of magnitude over the same sweep. Hence ϵ affects not only the numerical value of the curvature estimate but also the size of the virtual policy displacement on which the token-ranking proxy is evaluated. This introduces an additional sensitivity for online use: changing the probe scale changes the effective output-space regime of the diagnostic even when the underlying checkpoint and trajectories are fixed. The checkpoint-footprint signal used by OSOL does not require this additional finite-difference probe; it is measured directly from the log-probability change between realized adjacent checkpoints. Figure 9 therefore complements the accuracy and memory controls in Appendix A.4: the limitation of the Hessian route is not only computational cost, but also the need to choose a numerical perturbation scale whose effect propagates into the policy-space quantity being

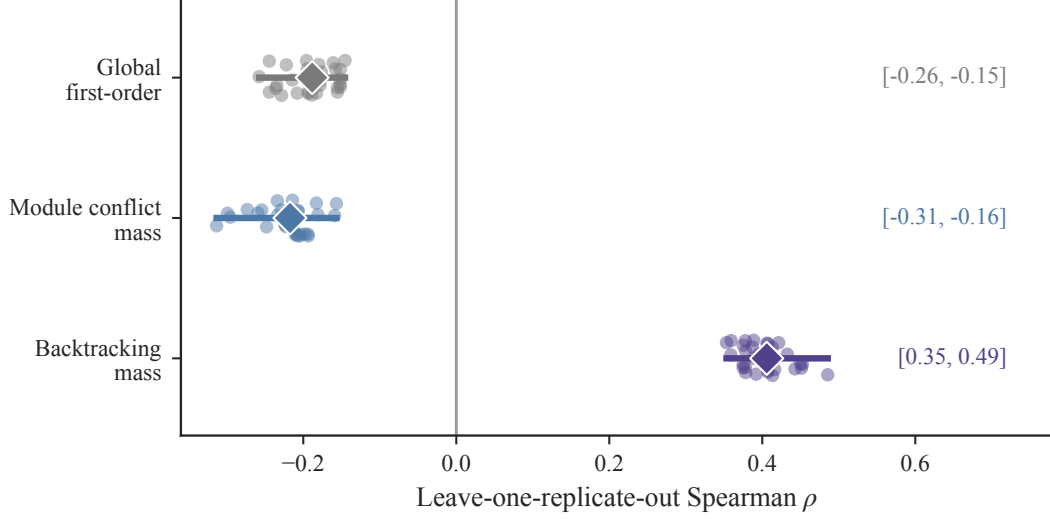


Figure 8: Leave-one-replicate stability of the Study I correlations. Each point recomputes Spearman ρ after removing one replicate; horizontal ranges summarize all 32 deletions. The sign and ordering of the diagnostics are preserved throughout.

ranked.

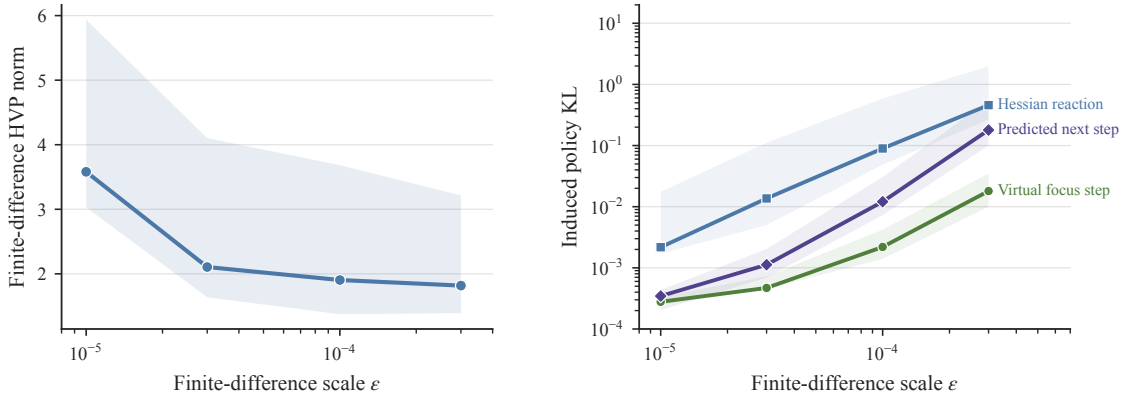


Figure 9: Sensitivity of the finite-difference Hessian proxy to the probe scale ϵ . Left: the estimated HVP norm varies with ϵ . Right: the induced policy KL of the virtual focus step, Hessian reaction, and predicted next step changes by orders of magnitude, showing that the probe scale also changes the output-space displacement being evaluated.

Taken together, these controls sharpen the mechanism evidence without changing the role assigned to each study in the main text. Study I shows that realized cross-step backtracking is a stronger *post-update diagnostic* of task damage than same-point gradient scores. Study II shows that finite-difference curvature introduces an additional probe-scale choice whose effect propagates into the output-space quantity used for token ranking. The online controller instead uses the preceding checkpoint footprint, which is already available before the next update and whose rebound-ranking ability is evaluated in Section 5.4. The results are therefore complementary: realized two-step backtracking characterizes the interaction that occurred, while the preceding footprint provides the pre-update information used to determine the intervention set.

C Output-Space Geometry

C.1 Notation and local regularity

Fix iteration k and the eligible focus coordinates $\mathcal{E}_k = \{(x_i, a_i)\}_{i=1}^N$. For an arbitrary action a at context x_i , define

$$\ell_{i,a}(\boldsymbol{\theta}) := \log \pi_{\boldsymbol{\theta}}(a \mid x_i), \quad \boldsymbol{\varphi}_{i,a} := \nabla \ell_{i,a}(\boldsymbol{\theta}_k). \quad (38)$$

For the realized action a_i , abbreviate

$$\ell_i(\boldsymbol{\theta}) := \ell_{i,a_i}(\boldsymbol{\theta}), \quad \boldsymbol{\varphi}_i := \boldsymbol{\varphi}_{i,a_i}, \quad \boldsymbol{\ell}_k(\boldsymbol{\theta}) := [\ell_1(\boldsymbol{\theta}), \dots, \ell_N(\boldsymbol{\theta})]^\top. \quad (39)$$

Collect the realized scores into

$$\boldsymbol{\Phi}_k := [\boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_N] \in \mathbb{R}^{P \times N}, \quad \mathbf{K}_k := \boldsymbol{\Phi}_k^\top \boldsymbol{\Phi}_k \in \mathbb{R}^{N \times N}. \quad (40)$$

Then

$$\mathbf{F}_k = \frac{1}{N} \boldsymbol{\Phi}_k \boldsymbol{\Phi}_k^\top, \quad \overline{\mathbf{F}}_k = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{a \sim \pi_{\boldsymbol{\theta}_k}(\cdot \mid x_i)} [\boldsymbol{\varphi}_{i,a} \boldsymbol{\varphi}_{i,a}^\top]. \quad (41)$$

We use the following local conditions, standard for Taylor and trust-region analyses [Nesterov, 2013, Nocedal and Wright, 2006].

1. The policy support is fixed in a neighborhood $\mathcal{N}_k$ of $\boldsymbol{\theta}_k$, and $\ell_{i,a}$ is three times continuously differentiable on $\mathcal{N}_k$.
2. There are constants $G, L, M < \infty$ such that, for every relevant (i, a) and $\boldsymbol{\theta} \in \mathcal{N}_k$,

$$\|\nabla \ell_{i,a}(\boldsymbol{\theta})\| \leq G, \quad \|\nabla^2 \ell_{i,a}(\boldsymbol{\theta})\|_{\text{op}} \leq L, \quad \|\nabla^3 \ell_{i,a}(\boldsymbol{\theta})\|_{\text{op}} \leq M. \quad (42)$$

3. The line segments joining $\boldsymbol{\theta}_{k-1}$, $\boldsymbol{\theta}_k$, $\boldsymbol{\theta}_{k+1}^0$, and the locally corrected endpoint remain in $\mathcal{N}_k$.
4. The adjacent displacements satisfy

$$\|\boldsymbol{\delta}_{k-1}\| \leq a\eta, \quad \|\boldsymbol{\delta}_k^0\| \leq b\eta, \quad \|\boldsymbol{\delta}_k^r\| \leq c_r\eta \quad (43)$$

for constants a, b, c_r independent of η .

The checkpoint identity in Appendix D is deterministic for the realized token coordinates and does not require independence across them. Sampling assumptions enter only when interpreting $\mathbf{F}_k$ as an estimator of $\overline{\mathbf{F}}_k$.

C.2 Relation to objective-Hessian perturbation analyses

Prior local-perturbation work studies damage to an earlier task objective after a later domain update. Around an approximately stationary earlier-task checkpoint, Taylor expansion makes the objective-Hessian quadratic term dominant [Yang et al., 2026a]; curvature-guided multi-domain optimization similarly motivates explicit or approximate curvature information [Liang et al., 2026]. These results explain why same-point gradient cosine can miss sequential interference, but their primary object is the self-quadratic sensitivity of one displacement under an objective Hessian.

Our analysis changes both the scalar function and the interaction being studied. We expand the task-conditioned output KL in Eq. (2), whose exact Hessian at the anchor is the Fisher matrix, and then evaluate the mixed output interaction between the preceding realized displacement and the current base displacement. Consequently, Eq. (6) is a bilinear second-order term in output geometry, not an identification of the complete objective Hessian with the Fisher. This output-space choice follows natural-gradient and trust-region policy optimization [Amari, 1998, Kakade, 2001, Schulman et al., 2015, Martens, 2020], while making the interaction observable through checkpoint log-probabilities.

C.3 Proof that task-conditioned output KL has Fisher Hessian

For one context x_i , write

$$D_i(\delta) := D_{\text{KL}}(\pi_{\theta_k}(\cdot | x_i) \parallel \pi_{\theta_k + \delta}(\cdot | x_i)). \quad (44)$$

Let $p_i(a) := \pi_{\theta_k}(a | x_i)$. Since the first distribution is fixed with respect to δ ,

$$D_i(\delta) = \sum_a p_i(a) \log p_i(a) - \sum_a p_i(a) \ell_{i,a}(\theta_k + \delta). \quad (45)$$

At $\delta = \mathbf{0}$, the two distributions coincide, so

$$D_i(\mathbf{0}) = 0. \quad (46)$$

Differentiating Eq. (45),

$$\nabla_{\delta} D_i(\delta) = - \sum_a p_i(a) \nabla \ell_{i,a}(\theta_k + \delta). \quad (47)$$

At the anchor,

$$\sum_a p_i(a) \varphi_{i,a} = \sum_a \pi_{\theta_k}(a | x_i) \nabla \log \pi_{\theta_k}(a | x_i) = \sum_a \nabla \pi_{\theta_k}(a | x_i) = \nabla 1 = \mathbf{0}. \quad (48)$$

Therefore

$$\nabla D_i(\mathbf{0}) = \mathbf{0}. \quad (49)$$

Differentiate the normalization identity once more. For each action,

$$\nabla^2 \pi_{\theta}(a | x_i) = \pi_{\theta}(a | x_i) [\nabla^2 \log \pi_{\theta}(a | x_i) + \nabla \log \pi_{\theta}(a | x_i) \nabla \log \pi_{\theta}(a | x_i)^{\top}]. \quad (50)$$

Summing over a gives zero because $\sum_a \pi_{\theta}(a | x_i) = 1$. Evaluated at θ_k ,

$$- \sum_a p_i(a) \nabla^2 \ell_{i,a}(\theta_k) = \sum_a p_i(a) \varphi_{i,a} \varphi_{i,a}^{\top}. \quad (51)$$

On the other hand, differentiating Eq. (47) gives

$$\nabla^2 D_i(\mathbf{0}) = - \sum_a p_i(a) \nabla^2 \ell_{i,a}(\theta_k). \quad (52)$$

Combining Eqs. (51) and (52),

$$\nabla^2 D_i(\mathbf{0}) = \mathbb{E}_{a \sim \pi_{\theta_k}(\cdot | x_i)} [\varphi_{i,a} \varphi_{i,a}^{\top}]. \quad (53)$$

Averaging over i proves Eq. (4). This is the standard Fisher–KL identity [Amari, 1998, Martens, 2020].

C.4 Explicit KL Taylor remainder and mixed derivative

Apply multivariate Taylor expansion to $\mathcal{D}_k$ at $\mathbf{0}$. Since its value and gradient vanish,

$$\mathcal{D}_k(\delta) = \frac{1}{2} \delta^{\top} \nabla^2 \mathcal{D}_k(\mathbf{0}) \delta + R_{\text{KL}}(\delta). \quad (54)$$

Under the third-derivative bound in Eq. (42), the integral or Lagrange remainder satisfies

$$|R_{\text{KL}}(\delta)| \leq \frac{M_{\mathcal{D}}}{6} \|\delta\|^3 \quad (55)$$

for a local constant $M_{\mathcal{D}}$. Substituting Eq. (4) into Eq. (54) proves the local expansion stated in Lemma 1. The cubic order follows from local Hessian Lipschitzness or, equivalently here, a locally bounded third derivative [Nesterov, 2013, Nocedal and Wright, 2006].

For two directions δ_1 and δ_2 , define $Q(s, t) := \mathcal{D}_k(s\delta_1 + t\delta_2)$. The chain rule gives

$$\left. \frac{\partial^2 Q}{\partial s \partial t} \right|_{s=t=0} = \delta_1^{\top} \overline{\mathbf{F}}_k \delta_2. \quad (56)$$

Equivalently, polarization of the Fisher quadratic form yields

$$2\delta_1^{\top} \overline{\mathbf{F}}_k \delta_2 = \|\delta_1 + \delta_2\|_{\overline{\mathbf{F}}_k}^2 - \|\delta_1\|_{\overline{\mathbf{F}}_k}^2 - \|\delta_2\|_{\overline{\mathbf{F}}_k}^2, \quad (57)$$

where $\|\delta\|_{\overline{\mathbf{F}}_k}^2 := \delta^{\top} \overline{\mathbf{F}}_k \delta$.

D Checkpoint Observability of Mixed Output Interaction

This section derives the preceding and current base footprints on the same realized token coordinates and proves that their product recovers the mixed output interaction under the empirical output Fisher up to cubic local error.

D.1 Taylor expansion of the preceding checkpoint footprint

Recall $\theta_{k-1} = \theta_k - \delta_{k-1}$. For each realized coordinate,

$$u_i = \ell_i(\theta_k) - \ell_i(\theta_k - \delta_{k-1}). \quad (58)$$

Taylor expansion of $\ell_i(\theta_k - \delta_{k-1})$ around θ_k gives

$$\ell_i(\theta_k - \delta_{k-1}) = \ell_i(\theta_k) - \varphi_i^\top \delta_{k-1} + \frac{1}{2} \delta_{k-1}^\top \nabla^2 \ell_i(\theta_k - t_i \delta_{k-1}) \delta_{k-1} \quad (59)$$

for some $t_i \in (0, 1)$. Hence

$$u_i = \varphi_i^\top \delta_{k-1} + e_i^u, \quad (60)$$

where

$$e_i^u := -\frac{1}{2} \delta_{k-1}^\top \nabla^2 \ell_i(\theta_k - t_i \delta_{k-1}) \delta_{k-1}, \quad |e_i^u| \leq \frac{L}{2} \|\delta_{k-1}\|^2. \quad (61)$$

In vector form,

$$\mathbf{u} = \Phi_k^\top \delta_{k-1} + \mathbf{e}^u. \quad (62)$$

D.2 Taylor expansion of the current base footprint

The current base update ends at $\theta_{k+1}^0 = \theta_k + \delta_k^0$. Therefore

$$d_i^0 = \ell_i(\theta_k + \delta_k^0) - \ell_i(\theta_k). \quad (63)$$

Taylor expansion around θ_k gives

$$d_i^0 = \varphi_i^\top \delta_k^0 + e_i^d, \quad (64)$$

with

$$e_i^d := \frac{1}{2} (\delta_k^0)^\top \nabla^2 \ell_i(\theta_k + t_i' \delta_k^0) \delta_k^0, \quad |e_i^d| \leq \frac{L}{2} \|\delta_k^0\|^2. \quad (65)$$

Thus

$$\mathbf{d}^0 = \Phi_k^\top \delta_k^0 + \mathbf{e}^d. \quad (66)$$

D.3 Proof of checkpoint token–Fisher duality

Multiplying Eqs. (60) and (64) coordinatewise gives

$$\begin{aligned} u_i d_i^0 &= (\varphi_i^\top \delta_{k-1}) (\varphi_i^\top \delta_k^0) \\ &\quad + (\varphi_i^\top \delta_{k-1}) e_i^d + (\varphi_i^\top \delta_k^0) e_i^u + e_i^u e_i^d. \end{aligned} \quad (67)$$

Averaging the leading term,

$$\frac{1}{N} \sum_{i=1}^N (\varphi_i^\top \delta_{k-1}) (\varphi_i^\top \delta_k^0) = \delta_{k-1}^\top \left(\frac{1}{N} \sum_{i=1}^N \varphi_i \varphi_i^\top \right) \delta_k^0 = \delta_{k-1}^\top \mathbf{F}_k \delta_k^0. \quad (68)$$

Using $|\varphi_i^\top \delta| \leq G \|\delta\|$ and Eqs. (61)–(65), the total error satisfies

$$\begin{aligned} \left| \frac{1}{N} \mathbf{u}^\top \mathbf{d}^0 - \delta_{k-1}^\top \mathbf{F}_k \delta_k^0 \right| &\leq \frac{GL}{2} \|\delta_{k-1}\| \|\delta_k^0\|^2 \\ &\quad + \frac{GL}{2} \|\delta_k^0\| \|\delta_{k-1}\|^2 \\ &\quad + \frac{L^2}{4} \|\delta_{k-1}\|^2 \|\delta_k^0\|^2. \end{aligned} \quad (69)$$

Under Eq. (43), the first two terms are $O(\eta^3)$ and the last is $O(\eta^4)$, proving Theorem 1. Equation (69) also states the finite-step approximation explicitly in addition to the asymptotic form.

D.4 Population and empirical Fisher

The checkpoint identity is deterministic in the realized empirical output metric $\mathbf{F}_k$, while the Hessian of the full action-distribution KL is $\bar{\mathbf{F}}_k$. This distinction is important because empirical Fisher matrices should not be identified with arbitrary objective Hessians [Kunstner et al., 2019].

If each realized a_i is sampled from the anchor policy conditionally on x_i , then

$$\mathbb{E}[\mathbf{F}_k \mid x_1, \dots, x_N] = \bar{\mathbf{F}}_k. \quad (70)$$

For fixed directions δ_1, δ_2 , define

$$Z_i := (\varphi_i^\top \delta_1)(\varphi_i^\top \delta_2). \quad (71)$$

Let

$$\bar{\mathbf{F}}_{k,i} := \mathbb{E}_{a \sim \pi_{\theta_k}(\cdot \mid x_i)}[\varphi_{i,a} \varphi_{i,a}^\top]. \quad (72)$$

Then

$$\delta_1^\top \mathbf{F}_k \delta_2 = \frac{1}{N} \sum_{i=1}^N Z_i, \quad \mathbb{E}[Z_i \mid x_i] = \delta_1^\top \bar{\mathbf{F}}_{k,i} \delta_2. \quad (73)$$

If $\|\varphi_i\| \leq G$, then

$$|Z_i| \leq G^2 \|\delta_1\| \|\delta_2\|. \quad (74)$$

Under conditional independence, Hoeffding’s inequality [Hoeffding, 1963] gives, with probability at least $1 - \delta$,

$$|\delta_1^\top (\mathbf{F}_k - \bar{\mathbf{F}}_k) \delta_2| \leq G^2 \|\delta_1\| \|\delta_2\| \sqrt{\frac{2 \log(2/\delta)}{N}}. \quad (75)$$

Combining Eqs. (69) and (75) yields

$$\frac{1}{N} \mathbf{u}^\top \mathbf{d}^0 = \delta_{k-1}^\top \bar{\mathbf{F}}_k \delta_k^0 + O(\eta^3) + O_p\left(\frac{\eta^2}{\sqrt{N}}\right). \quad (76)$$

For autoregressive trajectories, the deterministic empirical identity remains unchanged; only the optional population concentration argument is applied at the trajectory level or through an appropriate martingale bound.

D.5 Path cancellation and token-level backtracking

For arbitrary vectors $\mathbf{u}, \mathbf{d} \in \mathbb{R}^N$,

$$\|\mathbf{u} + \mathbf{d}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{d}\|^2 + 2\mathbf{u}^\top \mathbf{d}. \quad (77)$$

Rearranging proves Eq. (7). Its left-hand side is the sum of the two stepwise squared path lengths minus the squared net displacement, divided by two; it is positive when the two output motions exhibit net cancellation.

For the coordinatewise decomposition, define

$$\text{Align}(\mathbf{u}, \mathbf{d}) := \frac{1}{N} \sum_i [u_i d_i]_+, \quad \text{Rev}(\mathbf{u}, \mathbf{d}) := \frac{1}{N} \sum_i [-u_i d_i]_+. \quad (78)$$

Since every scalar x satisfies $x = [x]_+ - [-x]_+$,

$$\frac{1}{N} \mathbf{u}^\top \mathbf{d} = \text{Align}(\mathbf{u}, \mathbf{d}) - \text{Rev}(\mathbf{u}, \mathbf{d}), \quad \Gamma(\mathbf{u}, \mathbf{d}) = \text{Rev}(\mathbf{u}, \mathbf{d}) - \text{Align}(\mathbf{u}, \mathbf{d}). \quad (79)$$

Therefore

$$[\Gamma(\mathbf{u}, \mathbf{d})]_+ = [\text{Rev} - \text{Align}]_+ \leq \text{Rev}. \quad (80)$$

Token-level backtracking avoids cancellation between aligned and opposing coordinates. It decomposes into the two sign components

$$\text{Rev}(\mathbf{u}, \mathbf{d}) = \underbrace{\frac{1}{N} \sum_{i: u_i < 0} (-u_i)[d_i]_+}_{\mathcal{B}^-(\mathbf{u}, \mathbf{d})} + \underbrace{\frac{1}{N} \sum_{i: u_i > 0} u_i[-d_i]_+}_{\mathcal{B}^+(\mathbf{u}, \mathbf{d})}. \quad (81)$$

The first component is the negative-to-positive rebound controlled by OSOL.

E From Residual Coefficients to Backtracking Control

This section proves the control chain used in the main text: the constructed residual remains nonpositive after coefficient clipping, acts as a PPO likelihood-ratio penalty, induces a correction footprint through the token kernel, and contracts the targeted backtracking component.

E.1 Effective coefficient correction after clipping

The method constructs $r_i^{\text{OSOL}} \leq 0$ on $\mathcal{C}_k$ and $r_i^{\text{OSOL}} = 0$ outside it; the complete rank and scale construction is given in Appendix F.3. Set

$$\begin{aligned} A_i^0 &:= \text{clip}(A_i^{\text{base}}, -A_{\max}, A_{\max}), \\ \tilde{A}_i &:= \text{clip}(A_i^{\text{base}} + r_i^{\text{OSOL}}, -A_{\max}, A_{\max}), \quad \xi_i := \tilde{A}_i - A_i^0. \end{aligned} \quad (82)$$

The scalar clipping map is monotone nondecreasing. Hence

$$\xi_i \leq 0. \quad (83)$$

It is also 1-Lipschitz, so

$$|\xi_i| \leq |r_i^{\text{OSOL}}|. \quad (84)$$

Final coefficient clipping therefore never reverses the penalty direction and can only attenuate its magnitude. If $r_i^{\text{OSOL}} = 0$, then $\xi_i = 0$.

E.2 Proof of the history-conditioned PPO penalty

Define the token log-ratio and likelihood ratio

$$z_i(\theta) := \log \frac{\pi_{\theta}(a_i | x_i)}{\pi_{\theta_k}(a_i | x_i)}, \quad \omega_i(\theta) := e^{z_i(\theta)}. \quad (85)$$

At the anchor, $z_i = 0$ and $\omega_i = 1$, which lies in the interior of the PPO clipping interval. In a local neighborhood, the token losses therefore reduce to the unclipped likelihood-ratio form [Schulman et al., 2017]:

$$\mathcal{J}_i^0(z_i) = -e^{z_i} A_i^0, \quad \mathcal{J}_i^{\text{OSOL}}(z_i) = -e^{z_i} (A_i^0 + \xi_i). \quad (86)$$

Subtracting,

$$\mathcal{J}_i^{\text{OSOL}}(z_i) - \mathcal{J}_i^0(z_i) = -e^{z_i} \xi_i = e^{z_i} |\xi_i| \geq 0. \quad (87)$$

Differentiating with respect to z_i and evaluating at the anchor gives

$$\left. \frac{\partial(\mathcal{J}_i^{\text{OSOL}} - \mathcal{J}_i^0)}{\partial z_i} \right|_{z_i=0} = -\xi_i \geq 0, \quad (88)$$

which proves Lemma 2. Gradient descent therefore adds a penalty against increasing z_i on the selected support. The result is independent of the sign of A_i^0 because it concerns the incremental term induced by ξ_i .

E.3 Base update and correction footprint

Partition the response-token score matrix into focus and non-focus blocks, Φ_f and Φ_{nf} . In the local unclipped policy-gradient model [Sutton et al., 1999, Schulman et al., 2017], the base mixed update is

$$\delta_k^0 = \frac{\eta}{|\mathcal{R}|} (\Phi_f A_f^0 + \Phi_{\text{nf}} A_{\text{nf}}^0). \quad (89)$$

Let $\Phi_{\mathcal{E}}$ contain the scores of eligible focus coordinates. Their first-order base footprint is

$$\begin{aligned} d^0 &= \Phi_{\mathcal{E}}^{\top} \delta_k^0 + O(\eta^2) \\ &= \frac{\eta}{|\mathcal{R}|} (\Phi_{\mathcal{E}}^{\top} \Phi_f A_f^0 + \Phi_{\mathcal{E}}^{\top} \Phi_{\text{nf}} A_{\text{nf}}^0) + O(\eta^2). \end{aligned} \quad (90)$$

Thus $\mathbf{d}^0$ already includes the complete focus- and non-focus-domain contribution to the base update; the control argument does not remove or isolate non-focus contributions.

The effective OSOL correction is supported on eligible focus tokens:

$$\delta_k^r = \frac{\eta}{|\mathcal{R}|} \Phi_{\mathcal{E}} \xi. \quad (91)$$

Its correction footprint is

$$\mathbf{h} := \Phi_{\mathcal{E}}^\top \delta_k^r = \frac{\eta}{|\mathcal{R}|} \mathbf{K}_{\mathcal{E}} \xi, \quad \mathbf{K}_{\mathcal{E}} := \Phi_{\mathcal{E}}^\top \Phi_{\mathcal{E}}. \quad (92)$$

This is the standard first-order function-space motion induced by a Jacobian Gram matrix [Jacot et al., 2018]. Freezing the optimizer preconditioner at iteration k as a positive-semidefinite matrix P_k gives the local preconditioned response

$$\mathbf{h} = \frac{\eta}{|\mathcal{R}|} \Phi_{\mathcal{E}}^\top P_k \Phi_{\mathcal{E}} \xi. \quad (93)$$

The preconditioned token kernel remains positive semidefinite. For AdamW, P_k approximates the local change in the optimizer update with its moment state frozen at iteration k ; together with the unclipped PPO model at the anchor, this captures the first-order effect of the correction rather than the full clipped PPO–AdamW trajectory.

E.4 Residual-to-output sign transfer

From Eq. (92), for $i \in \mathcal{C}_k$,

$$h_i = \frac{\eta}{|\mathcal{R}|} \left[(\mathbf{K}_{\mathcal{E}})_{ii} \xi_i + \sum_{j \in \mathcal{C}_k, j \neq i} (\mathbf{K}_{\mathcal{E}})_{ij} \xi_j \right]. \quad (94)$$

A sufficient local self-response condition is

$$(\mathbf{K}_{\mathcal{E}})_{ii} |\xi_i| \geq \left| \sum_{j \in \mathcal{C}_k, j \neq i} (\mathbf{K}_{\mathcal{E}})_{ij} \xi_j \right|. \quad (95)$$

Since $(\mathbf{K}_{\mathcal{E}})_{ii} = \|\varphi_i\|^2 \geq 0$ and $\xi_i \leq 0$, the diagonal term is nonpositive. Under Eq. (95),

$$\begin{aligned} (\mathbf{K}_{\mathcal{E}})_{ii} \xi_i + \sum_{j \neq i} (\mathbf{K}_{\mathcal{E}})_{ij} \xi_j &\leq -(\mathbf{K}_{\mathcal{E}})_{ii} |\xi_i| + \left| \sum_{j \neq i} (\mathbf{K}_{\mathcal{E}})_{ij} \xi_j \right| \\ &\leq 0, \end{aligned} \quad (96)$$

which proves Lemma 3. The condition holds automatically for a diagonal token kernel and under weighted diagonal dominance on the selected coordinates.

Coordinatewise nonpositivity is sufficient but not necessary. Using

$$[x + y]_+ - [x]_+ = \int_0^1 \mathbf{1}\{x + ty > 0\} y \, dt \quad (97)$$

for almost every (x, y) , the exact first-order backtracking change is

$$\begin{aligned} \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0 + \mathbf{h}) - \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0) \\ = \frac{1}{N} \int_0^1 \sum_{i \in \mathcal{C}_k} (-u_i) \mathbf{1}\{d_i^0 + th_i > 0\} h_i \, dt. \end{aligned} \quad (98)$$

Hence the weaker aggregate condition

$$\sum_{i \in \mathcal{C}_k} (-u_i) \mathbf{1}\{d_i^0 + th_i > 0\} h_i \leq 0, \quad t \in [0, 1], \quad (99)$$

is sufficient for contraction even if a small number of individual h_i are positive because of cross-token coupling.

E.5 Proof of first-order backtracking contraction

For $i \in \mathcal{C}_k$, $u_i < 0$. Under $h_i \leq 0$, monotonicity of the positive-part function gives

$$[d_i^0 + h_i]_+ \leq [d_i^0]_+. \quad (100)$$

Multiplying by $-u_i > 0$, summing, and dividing by N yields

$$\mathcal{B}^-(\mathbf{u}, \mathbf{d}^0 + \mathbf{h}) \leq \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0). \quad (101)$$

The nonnegative difference is exactly the control gain in Eq. (12). For the selected-coordinate signed interaction,

$$\begin{aligned} & \frac{1}{N} \sum_{i \in \mathcal{C}_k} u_i (d_i^0 + h_i) - \frac{1}{N} \sum_{i \in \mathcal{C}_k} u_i d_i^0 \\ &= \frac{1}{N} \sum_{i \in \mathcal{C}_k} u_i h_i \geq 0, \end{aligned} \quad (102)$$

because both factors are nonpositive. Strict inequality holds whenever $u_i < 0$ and $h_i < 0$ on at least one selected coordinate.

E.6 Finite-step control gain

The local proof uses first-order footprints. The corrected footprint is

$$\mathbf{d}^{\text{OSOL}} := \ell_k(\boldsymbol{\theta}_k + \boldsymbol{\delta}_k^0 + \boldsymbol{\delta}_k^r) - \ell_k(\boldsymbol{\theta}_k). \quad (103)$$

Taylor expansion around $\boldsymbol{\theta}_k$ gives

$$\mathbf{d}^{\text{OSOL}} = \mathbf{d}^0 + \mathbf{h} + \mathbf{e}^r, \quad \|\mathbf{e}^r\|_\infty = O(\eta^2). \quad (104)$$

Since $x \mapsto [x]_+$ is 1-Lipschitz,

$$\begin{aligned} & |\mathcal{B}^-(\mathbf{u}, \mathbf{d}^{\text{OSOL}}) - \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0 + \mathbf{h})| \\ & \leq \frac{1}{N} \sum_{i \in \mathcal{C}_k} (-u_i) |e_i^r| \leq \frac{\|\mathbf{u}\|_1}{N} \|\mathbf{e}^r\|_\infty = O(\eta^3), \end{aligned} \quad (105)$$

because $\|\mathbf{u}\|_1/N = O(\eta)$. Therefore

$$\begin{aligned} \mathcal{B}^-(\mathbf{u}, \mathbf{d}^{\text{OSOL}}) & \leq \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0 + \mathbf{h}) + O(\eta^3) \\ & = \mathcal{B}^-(\mathbf{u}, \mathbf{d}^0) - \Delta_k^{\text{ctrl}} + O(\eta^3), \end{aligned} \quad (106)$$

which proves Eq. (13). This form retains the controller's first-order reduction explicitly and separates it from the cubic deviation incurred by observing a finite checkpoint step.

F Design Principles and Code-Faithful Construction

The preceding sections establish how a nonpositive effective coefficient becomes a backtracking-reducing output correction. This section derives the three design choices used to construct that coefficient: penalty-only support, rank-based allocation, and batchwise scale normalization.

F.1 Why OSOL uses the penalty-only support

The full tokenwise backtracking statistic is symmetric in the sign patterns $u_i < 0, d_i > 0$ and $u_i > 0, d_i < 0$. A symmetric controller would apply a nonpositive correction after negative drift and a nonnegative correction after positive drift. OSOL restricts the correction to the negative-drift candidate set because a nonpositive correction has a

monotone backtracking-reduction property there and does not introduce a positive reward based only on checkpoint history.

For an arbitrary nonpositive output-space correction $v_i \leq 0$, define

$$b_i(v_i) := [-u_i(d_i + v_i)]_+. \quad (107)$$

If $u_i < 0$,

$$b_i(v_i) = (-u_i)[d_i + v_i]_+ \leq (-u_i)[d_i]_+ = b_i(0). \quad (108)$$

If $u_i > 0$,

$$b_i(v_i) = u_i[-d_i - v_i]_+ \geq u_i[-d_i]_+ = b_i(0). \quad (109)$$

Thus a nonpositive correction is backtracking-reducing on $\{u_i < 0\}$ but can increase backtracking on $\{u_i > 0\}$.

For any $\mathbf{v} \preceq 0$, define its negative-drift projection

$$\mathbf{v}^- := \begin{cases} \mathbf{v}, & u_i < 0, \\ 0, & u_i \geq 0. \end{cases} \quad (110)$$

Applying Eqs. (108)–(109) coordinatewise gives

$$\text{Rev}(\mathbf{u}, \mathbf{d} + \mathbf{v}^-) \leq \text{Rev}(\mathbf{u}, \mathbf{d} + \mathbf{v}). \quad (111)$$

This identifies $\{u_i < 0\}$ as the support on which a nonpositive correction cannot increase backtracking; Lemma 3 connects this support to the implemented coefficient correction.

Suppressing the complementary pattern $u_i > 0, d_i < 0$ would require a positive coefficient correction $\xi_i^+ > 0$. At the PPO anchor,

$$\mathcal{J}_i^+ - \mathcal{J}_i^0 = -e^{z_i} \xi_i^+, \quad \left. \frac{\partial(\mathcal{J}_i^+ - \mathcal{J}_i^0)}{\partial z_i} \right|_{z_i=0} = -\xi_i^+ < 0, \quad (112)$$

so gradient descent treats the historical signal as an additional reward-like incentive. This changes the role of history in four ways.

Historical-drift self-reinforcement. A preceding positive movement would itself trigger an additional positive coefficient, making the same direction more likely to continue even when the new rollout provides no confirming reward evidence. A transient checkpoint displacement can thereby become a persistent optimization preference.

Positive feedback on sampling noise. The measured drift contains finite-rollout, optimizer, and checkpoint variability. A positive residual maps a noisy positive realization into another positive update, creating a feedback loop that amplifies rather than damps the historical fluctuation.

Persistent elevation of incorrect tokens. A token probability can rise because of incidental generalization, cross-token coupling, or an update that benefits another domain. Historical sign alone does not certify correctness; rewarding every previous increase can repeatedly elevate a token whose current reward evidence is weak or adverse.

Conflict with the current GRPO advantage. When $A_i^{\text{base}} < 0$, a positive history residual cancels part of the current reward-grounded negative coefficient and can even reverse its sign. The update would then prefer historical motion over the latest within-group reward comparison.

This penalty-only design avoids introducing a new positive incentive solely from checkpoint history. It can attenuate a likelihood increase, but it cannot create a positive incentive absent from the current GRPO coefficient. This asymmetry is therefore a control design rather than an omission: OSOL uses the candidate set with a monotone backtracking-reduction guarantee while preserving the current reward as the sole source of positive reinforcement.

F.2 Rank-optimal local allocation

Write a nonpositive output correction as $v_i = -e_i$ with $e_i \geq 0$. For $u_i < 0$, the exact reduction in scalar backtracking is

$$\begin{aligned} [-u_i d_i]_+ - [-u_i(d_i - e_i)]_+ &= (-u_i)([d_i]_+ - [d_i - e_i]_+) \\ &= (-u_i) \min\{e_i, [d_i]_+\}. \end{aligned} \quad (113)$$

When $d_i > 0$ and $0 \leq e_i \leq d_i$, the reduction is linear:

$$[-u_i d_i]_+ - [-u_i(d_i - e_i)]_+ = (-u_i)e_i, \quad (114)$$

so the marginal reduction per unit penalty is $-u_i$.

Proposition 2 (Rank-optimal local allocation). *Consider active rebound coordinates with $u_i < 0$ and $d_i > 0$. Let $\{e_i\}$ be a fixed multiset of unsaturated penalty levels satisfying $0 \leq e_i \leq \min_j d_j$, so every level is feasible on every active coordinate. The total local reduction*

$$\Delta \mathcal{B}^- = \frac{1}{N} \sum_i (-u_i) e_i \quad (115)$$

is maximized by assigning larger e_i to more negative u_i .

Proof. Let $-u_i \geq -u_j$ but $e_i < e_j$. Swapping the two penalty levels changes the objective by

$$\frac{1}{N} [(-u_i)e_j + (-u_j)e_i - (-u_i)e_i - (-u_j)e_j] = \frac{1}{N} ((-u_i) - (-u_j))(e_j - e_i) \geq 0. \quad (116)$$

Repeatedly eliminating such inversions produces a co-monotone assignment without decreasing the reduction. Hence stronger penalties should be assigned to more negative preceding drift. $\square$

Proposition 2 supplies the allocation principle behind Eq. (16). The quantile map implements the optimal ordering without using raw drift magnitude, is robust to isolated extremes, and separates relative allocation from the common scale chosen by τ . Under the diagonal self-response approximation $h_i \simeq \alpha_i \xi_i$ with positive, locally comparable α_i , the same ordering transfers from coefficient penalties to the correction footprint.

F.3 Code-faithful residual construction

The base coefficient is

$$A_i^{\text{base}} = c_i A_i^{\text{g}}, \quad c_i = \frac{w_{\text{src}(i)}}{Z_k}, \quad (117)$$

where Z_k is given by Eq. (14). The history residual is not multiplied by c_i again. The eligible set $\mathcal{E}_k$ contains response-token coordinates from the current focus domain whose trajectory-level GRPO advantage is nonzero, and

$$\mathcal{C}_k = \{i \in \mathcal{E}_k : u_i < 0\}. \quad (118)$$

For $n_k := |\mathcal{C}_k| > 0$, sort candidate drifts in ascending order and assign

$$p_i = \frac{\text{rank}_0(u_i) + 1/2}{n_k}. \quad (119)$$

The clipped affine profile is

$$\bar{r}_i = r_{\min} + \frac{\text{clip}(p_i, p_{\text{lo}}, p_{\text{hi}}) - p_{\text{lo}}}{p_{\text{hi}} - p_{\text{lo}}} (r_{\max} - r_{\min}), \quad (120)$$

with $\bar{r}_i = 0$ off $\mathcal{C}_k$. Since ascending rank is monotone,

$$u_i \leq u_j < 0 \implies \bar{r}_i \leq \bar{r}_j \leq 0. \quad (121)$$

For $(p_{\text{lo}}, p_{\text{hi}}, r_{\min}, r_{\max}) = (0.1, 1, -1, 0)$,

$$\bar{r}_i = -1 + \frac{\text{clip}(p_i, 0.1, 1) - 0.1}{0.9}. \quad (122)$$

The implementation retains an explicit global gate $g_k \geq 0$ and focus mask $\lambda_i \in [0, 1]$:

$$q_i = g_k \lambda_i \bar{r}_i. \quad (123)$$

In the standard configuration, $g_k = 1$ and $\lambda_i = 1$ on the eligible focus coordinates, so Eq. (123) reduces to the conceptual profile in Eq. (17). The explicit terms are retained in code to support scheduling and masking without changing the core method.

Policy objective. After forming $\tilde{A}_i$ in Eq. (20), define

$$\omega_i(\boldsymbol{\theta}) := \frac{\pi_{\boldsymbol{\theta}}(a_i \mid x_i)}{\pi_{\boldsymbol{\theta}_{\text{old}}}(a_i \mid x_i)}. \quad (124)$$

The controller uses the unchanged clipped surrogate [Schulman et al., 2017]:

$$\mathcal{L}_{\text{OSOL}}(\boldsymbol{\theta}) = -\mathbb{E} \left[\frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} \min \left(\omega_i(\boldsymbol{\theta}) \tilde{A}_i, \text{clip}(\omega_i(\boldsymbol{\theta}), 1 - \epsilon_c, 1 + \epsilon_c) \tilde{A}_i \right) \right]. \quad (125)$$

No auxiliary objective is introduced; OSOL changes only the coefficient supplied to the existing policy update.

F.4 Adaptive scale identity and degenerate cases

Let

$$\kappa_k = \tau \frac{\sigma_A}{\sigma_q}, \quad \sigma_A := \text{std}_{\mathcal{E}_k}(A^{\text{base}}), \quad \sigma_q := \text{std}_{\mathcal{E}_k}(q). \quad (126)$$

When σ_A or σ_q is nonfinite, or σ_q falls below the implementation threshold, the method sets $\kappa_k = 0$. Otherwise,

$$r_i^{\text{OSOL}} = \kappa_k q_i = \kappa_k g_k \lambda_i \bar{r}_i \leq 0, \quad (127)$$

and homogeneity of standard deviation gives

$$\text{std}_{\mathcal{E}_k}(r^{\text{OSOL}}) = \kappa_k \text{std}_{\mathcal{E}_k}(q) = \tau \text{std}_{\mathcal{E}_k}(A^{\text{base}}). \quad (128)$$

Thus

$$\frac{\text{std}_{\mathcal{E}_k}(r^{\text{OSOL}})}{\text{std}_{\mathcal{E}_k}(A^{\text{base}})} = \tau. \quad (129)$$

The identity is a batchwise scale normalization: τ fixes residual dispersion relative to base-advantage dispersion. Because noncandidate zeros remain in the eligible-set statistic, a sparse candidate set concentrates that dispersion on fewer coordinates, whereas a dense set distributes it more broadly. If $\mathcal{C}_k = \emptyset$ or the scale statistics are degenerate, $\kappa_k = 0$ and OSOL reduces exactly to the base mixed-domain objective.

F.5 Main-Training Reproducibility Details

Table 4 summarizes the training, generation, and OSOL-specific settings used in the main experiments. Let $\mathcal{D}$ denote the four training domains. At trainer step k , the focus domain is sampled uniformly from the domains other than the preceding focus,

$$f_k \sim \text{Unif}(\mathcal{D} \setminus \{f_{k-1}\}). \quad (130)$$

The no-repeat constraint prevents one domain from leading two consecutive updates, while random selection avoids imposing a fixed domain order. All four domains remain in the mixed-domain batch; the sampled focus domain determines the asymmetric base weight and the response tokens eligible for history correction. Each trainer step contains 128 prompts and eight sampled responses per prompt, yielding 1,024 rollout trajectories before optimization. Unless otherwise specified, the optimization and rollout settings follow the standard GRPO configuration in verl v0.5.0.

The preceding checkpoint is used only to rescore the same realized focus-domain token coordinates sampled by the current policy. All checkpoint-derived quantities are treated as fixed coefficients during optimization, and gradients are not propagated through θ_{k-1} . The resulting history residual is added to the task-weighted GRPO coefficient before applying the standard clipped policy objective.

Table 4: Main-training and generation settings. The prompt batch size is counted before rollout expansion.

Setting	Configuration
Model backbones	Qwen3-30B-A3B and Qwen3-8B-Base
Training domains	Mathematics, code generation, instruction following, and open-ended Chat
Training examples	7.5K examples per domain
Focus-domain schedule	Uniform random sampling from $\mathcal{D} \setminus \{f_{k-1}\}$; consecutive steps cannot use the same focus domain
Prompt batch size	128 prompts per trainer step
Rollouts per prompt	8
Rollout trajectories per step	1,024
Training duration	15 epochs
Optimizer	AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01
Learning rate	1×10^{-6} with a constant schedule and no warmup
PPO epochs	1 per rollout batch
PPO clip ratio	0.2 (symmetric clipping to $[0.8, 1.2]$)
Sampling temperature	1.0
Top- p / Top- k	1.0 / -1 (no nucleus or top- k truncation)
Maximum response length	8,192 tokens
Chat reward model	Skywork-Reward-Llama-3.1-8B-v0.2, following the Chat reward configuration used by MGS
History checkpoint	The frozen preceding checkpoint θ_{k-1}
Eligible history tokens	Focus-domain response tokens with nonzero trajectory-level GRPO advantage
Negative-drift candidates	Tokens satisfying $u_i < 0$
Rank-profile parameters	$(p_{lo}, p_{hi}, r_{min}, r_{max}) = (0.1, 1, -1, 0)$
Residual scale	$\tau = 0.03$
Final coefficient range	$[-3, 3]$